

STATS141XP_Final_Project



2025-05-23

```
# load package
library(ggplot2)
library(dplyr)
library(tidyr)

# for model
library(glmnet)
library(dplyr)
library(purrr)
library(corrplot)
library(nnet)
```

Quick Data Cleaning

```
# removing unnecessary columns
df <- read.csv("updated_filtered_dataset.csv")
df <- df[, !names(df) %in% c("X.2", "X.1", "X", "RespondentID", "yearBorn", "date")] %>%
  filter(function. != "NR")
```

Focusing on psychometric traits

```
# Keep only psychometric traits + outcome
# Filter to top 10 most common job functions, excluding NR
df_traits <- df %>%
  filter(function. %in% (
    count(., function.) %>%
      slice_max(n, n = 10) %>%
      pull(function.)
  )) %>%
  select(starts_with("scaleA_"), starts_with("scaleD_"), function.) %>%
  na.omit() %>%
  mutate(function. = as.factor(function.))

head(df_traits)
```

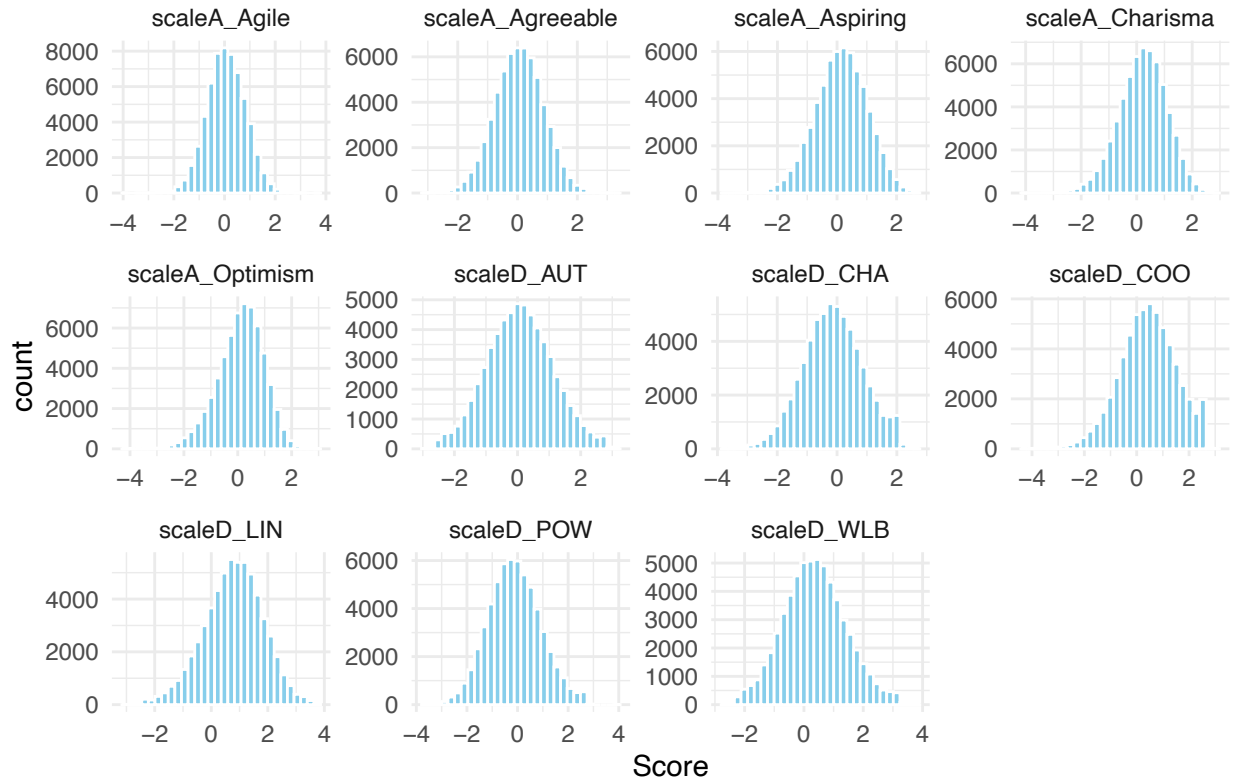
```
##   scaleA_Agile scaleA_Charisma scaleA_Aspiring scaleA_Agreeable scaleA_Optimism
## 1    0.7577258    0.1277281    -0.3012949    1.3738894    0.05823041
```

```
## 2    1.4630323    -0.4122419     0.1481820     1.5129885    -1.82403226
## 3    -0.3362857    -1.0686037    -0.3976152    -0.2017604    -0.40197235
## 4     0.4794793     0.8465945    -0.5740415    -0.1209470     0.40709447
## 5     0.9049424     1.1015668     1.2628963     1.2090530     1.22999539
## 6     0.9983525     1.7132005     1.5852949     0.2898318     1.08855300
##   scaled_WLB scaled_CHA scaled_C00 scaled_AUT scaled_POW scaled_LIN
## 1  0.5173929 -0.02991162  1.4244869 -0.01204274  0.3002343 -0.1960117
## 2  1.6579237  0.64876190  0.7982065 -0.28591261 -2.0535629 -2.4322007
## 3  1.4757888 -0.75131945 -0.8120493  0.46510334 -1.4410138  0.1714628
## 4  0.2603217 -0.38082418  0.9995238 -0.64412876 -0.5545732  1.1366830
## 5 -0.5899996  0.76602844  2.4156309  2.41014748  2.5994732 -1.6185748
## 6 -0.1832266 -1.24327053  1.2800712  2.03138146  1.5141974 -0.4218981
##           function.
## 1           Engineering
## 2           Engineering
## 3           Engineering
## 4      Software & Systems
## 5 Executive & General Management
## 6 Executive & General Management
```

Distribution check

```
df_traits %>%
  pivot_longer(-function., names_to = "Trait", values_to = "Score") %>%
  ggplot(aes(x = Score)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "white") +
  facet_wrap(~ Trait, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Psychometric Trait Scores")
```

Distribution of Psychometric Trait Scores

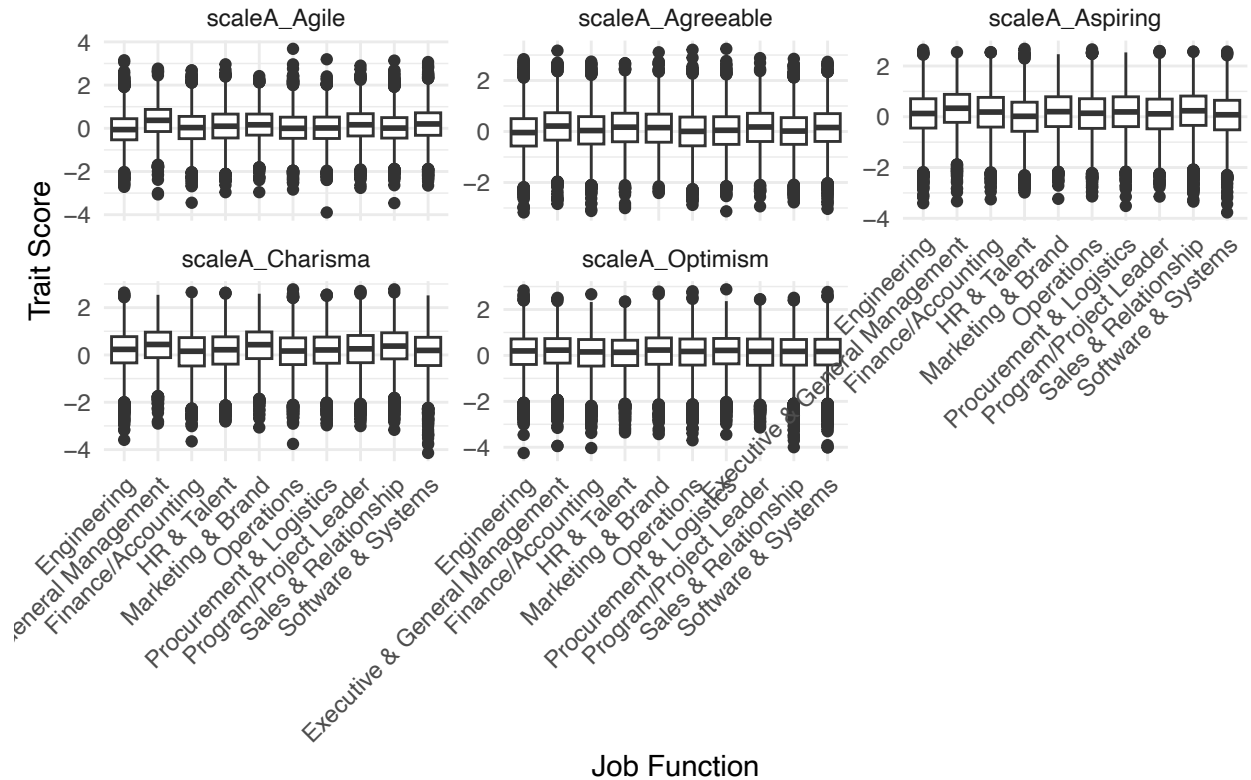


Boxplots Plots by Job Function

```
# Reshape to long format for selected traits
df_traits_long <- df_traits %>%
  pivot_longer(cols = starts_with("scaleA_"), names_to = "Trait", values_to = "Score")

ggplot(df_traits_long, aes(x = function., y = Score)) +
  geom_boxplot() +
  facet_wrap(~ Trait, scales = "free_y") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Boxplots of A-Traits by Job Function", x = "Job Function", y = "Trait Score")
```

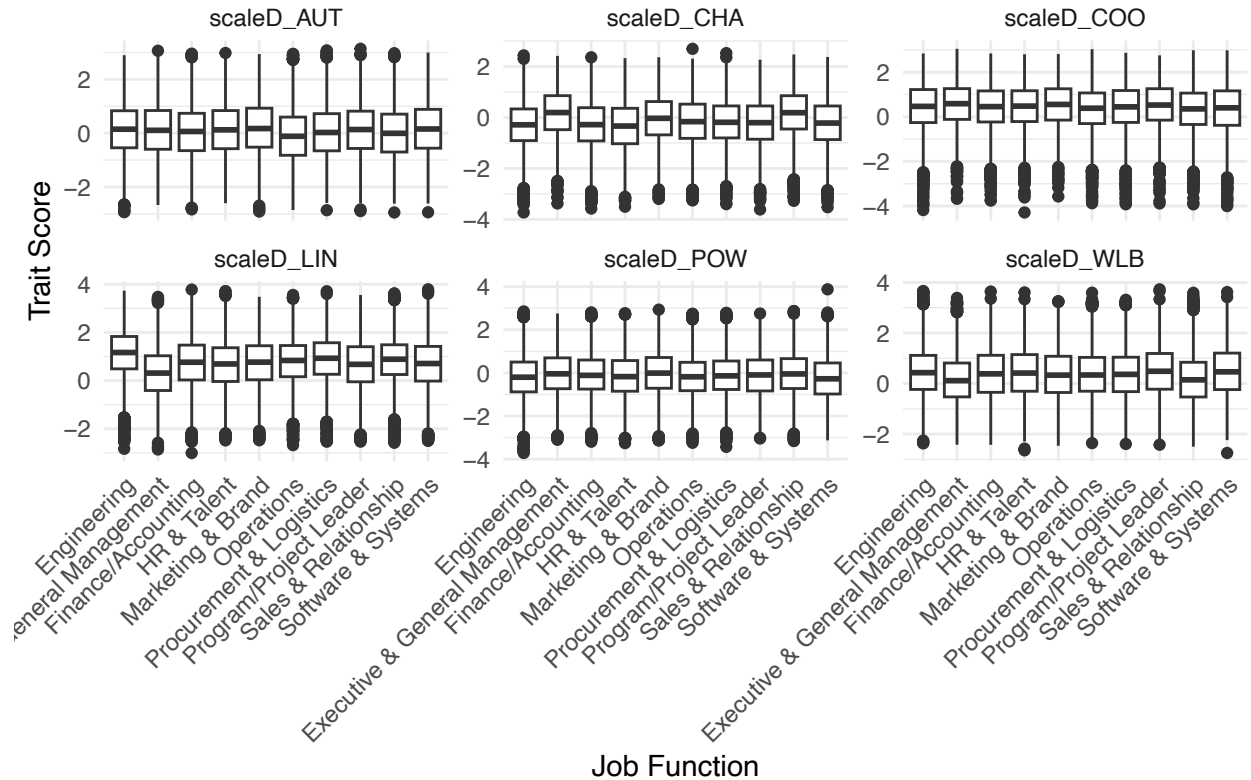
Boxplots of A-Traits by Job Function



```
# Reshape to long format for selected traits
df_traits_long <- df_traits %>%
  pivot_longer(cols = starts_with("scaleD_"), names_to = "Trait", values_to = "Score")

ggplot(df_traits_long, aes(x = function., y = Score)) +
  geom_boxplot() +
  facet_wrap(~ Trait, scales = "free_y") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Boxplots of D-Traits by Job Function", x = "Job Function", y = "Trait Score")
```

Boxplots of D–Traits by Job Function



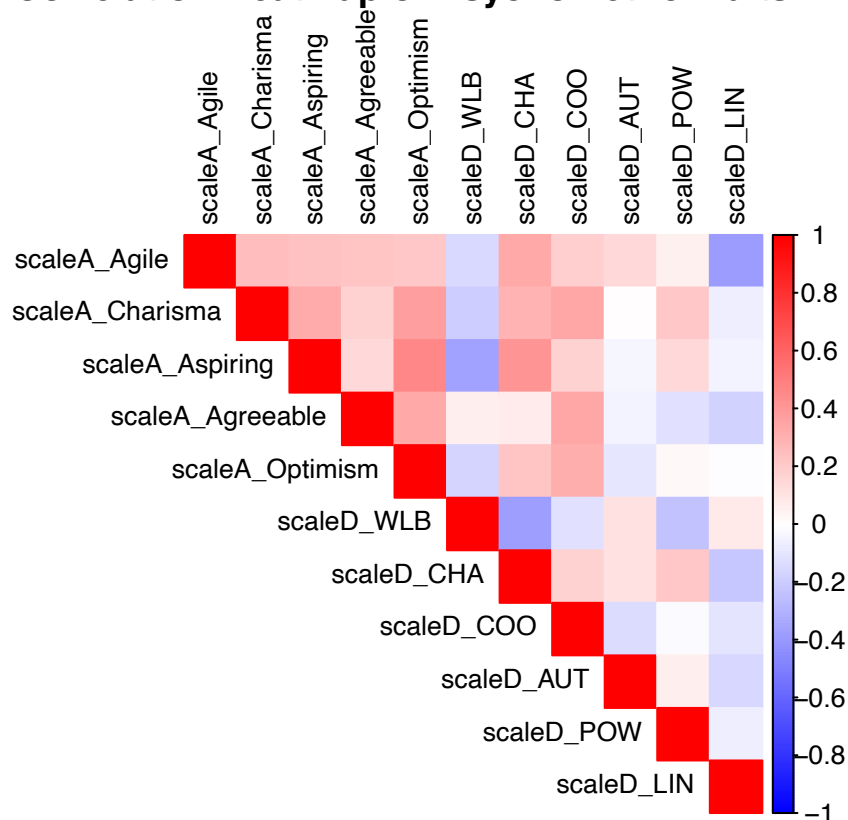
Correlation Heat Map

```
# Select only psychometric traits (A and D scales)
traits <- df_traits %>% select(starts_with("scaleA_"), starts_with("scaleD_"))

# Compute correlation matrix
cor_matrix <- cor(traits, use = "complete.obs")

# Plot heatmap
corrplot(cor_matrix, method = "color", type = "upper",
         tl.col = "black", tl.cex = 0.8,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         title = "Correlation Heatmap of Psychometric Traits", mar = c(0,0,1,0))
```

Correlation Heatmap of Psychometric Traits



Fit the Model

```
set.seed(123)

X <- as.matrix(df_traits %>% select(-function.))
y <- df_traits$function.

fit <- cv.glmnet(X, y, family = "multinomial", alpha = 1)

# View non-zero coefficients at best lambda
coefs <- coef(fit, s = "lambda.min")
coefs
```

```
## $Engineering
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)  0.257448058
## scaleA_Agile -0.053127531
## scaleA_Charisma 0.003732178
## scaleA_Aspiring 0.046430495
## scaleA_Agreeable -0.200995949
## scaleA_Optimism 0.085550307
## scaleD_WLB .
## scaleD_CHA -0.055544663
## scaleD_COO 0.074757855
## scaleD_AUT 0.115763520
## scaleD_POW -0.067702272
```

```

## scaled_LIN      0.357462590
##
## $'Executive & General Management'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.07878349
## scaleA_Agile   0.19830877
## scaleA_Charisma 0.17148815
## scaleA_Aspiring 0.10785028
## scaleA_Agreeable 0.03366208
## scaleA_Optimism -0.11958533
## scaled_WLB     -0.09355310
## scaled_CHA      0.18192776
## scaled_COO      -0.05231805
## scaled_AUT      -0.07839446
## scaled_POW      -0.01204233
## scaled_LIN      -0.29431920
##
## $'Finance/Accounting'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.003136738
## scaleA_Agile   -0.044473379
## scaleA_Charisma -0.148035407
## scaleA_Aspiring 0.179486109
## scaleA_Agreeable -0.094304501
## scaleA_Optimism .
## scaled_WLB     -0.007387510
## scaled_CHA      -0.135907588
## scaled_COO      0.035536358
## scaled_AUT      -0.017569180
## scaled_POW      0.051480077
## scaled_LIN      -0.045069275
##
## $'HR & Talent'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.025645293
## scaleA_Agile    0.044473379
## scaleA_Charisma .
## scaleA_Aspiring -0.117628465
## scaleA_Agreeable 0.083554006
## scaleA_Optimism .
## scaled_WLB     -0.063812333
## scaled_CHA      -0.187609950
## scaled_COO      0.009737875
## scaled_AUT      0.042668217
## scaled_POW      0.012042328
## scaled_LIN      -0.083596757
##
## $'Marketing & Brand'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.6421162457

```

```

## scaleA_Agile      0.0535929127
## scaleA_Charisma   0.1758814349
## scaleA_Aspiring   -0.0006469685
## scaleA_Agreeable  0.0259221899
## scaleA_Optimism    -0.0232942315
## scaled_WLB        0.0173997737
## scaled_CHA        0.0954748791
## scaled_COO        0.0115209380
## scaled_AUT        0.0705902464
## scaled_POW        0.0668205947
## scaled_LIN        0.0525330238
##
## $Operations
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   4.916918e-01
## scaleA_Agile   -5.910664e-02
## scaleA_Charisma -5.915677e-02
## scaleA_Aspiring .
## scaleA_Agreeable -1.567906e-01
## scaleA_Optimism  7.319268e-02
## scaled_WLB      1.410683e-05
## scaled_CHA      1.052550e-01
## scaled_COO      -8.759498e-02
## scaled_AUT      -1.996626e-01
## scaled_POW      -5.585215e-02
## scaled_LIN      -2.678970e-03
##
## $'Procurement & Logistics'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.241549257
## scaleA_Agile   -0.045132993
## scaleA_Charisma -0.041397630
## scaleA_Aspiring  0.088822027
## scaleA_Agreeable -0.091179813
## scaleA_Optimism  0.069661227
## scaled_WLB      .
## scaled_CHA      0.004361985
## scaled_COO      -0.009737875
## scaled_AUT      -0.022666056
## scaled_POW      0.012984885
## scaled_LIN      0.124838802
##
## $'Program/Project Leader'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)   -0.29628290
## scaleA_Agile    0.10457525
## scaleA_Charisma .
## scaleA_Aspiring .
## scaleA_Agreeable 0.02668541
## scaleA_Optimism -0.01344954
## scaled_WLB      0.07591896

```



```

## scaled_CHA      -0.04910945
## scaled_COO      0.04305230
## scaled_AUT      0.01756918
## scaled_POW      0.03826732
## scaled_LIN      -0.04886170
##
## $'Sales & Relationship'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)  0.53169394
## scaleA_Agile -0.23093632
## scaleA_Charisma 0.25500350
## scaleA_Aspiring -0.04586601
## scaleA_Agreeable -0.02592219
## scaleA_Optimism -0.09682559
## scaled_WLB    -0.10723443
## scaled_CHA     0.46235698
## scaled_COO    -0.19797501
## scaled_AUT    -0.11048987
## scaled_POW    -0.03845538
## scaled_LIN     0.12123062
##
## $'Software & Systems'
## 12 x 1 sparse Matrix of class "dgCMatrix"
##              1
## (Intercept)  0.006680138
## scaleA_Agile  0.257160348
## scaleA_Charisma -0.099727658
## scaleA_Aspiring -0.073902069
## scaleA_Agreeable 0.040583886
## scaleA_Optimism 0.051681672
## scaled_WLB     0.030375643
## scaled_CHA    -0.004361985
## scaled_COO    -0.102405998
## scaled_AUT     0.017724378
## scaled_POW    -0.080948871
## scaled_LIN     0.002678970

```

```

# Convert the list of sparse matrices to a tidy dataframe
coef_df <- map_dfr(names(coefs), function(name) {
  mat <- as.matrix(coefs[[name]])
  data.frame(
    JobFunction = name,
    Trait = rownames(mat),
    Coefficient = mat[, 1],
    row.names = NULL
  )
}) %>% filter(Trait != "(Intercept)")

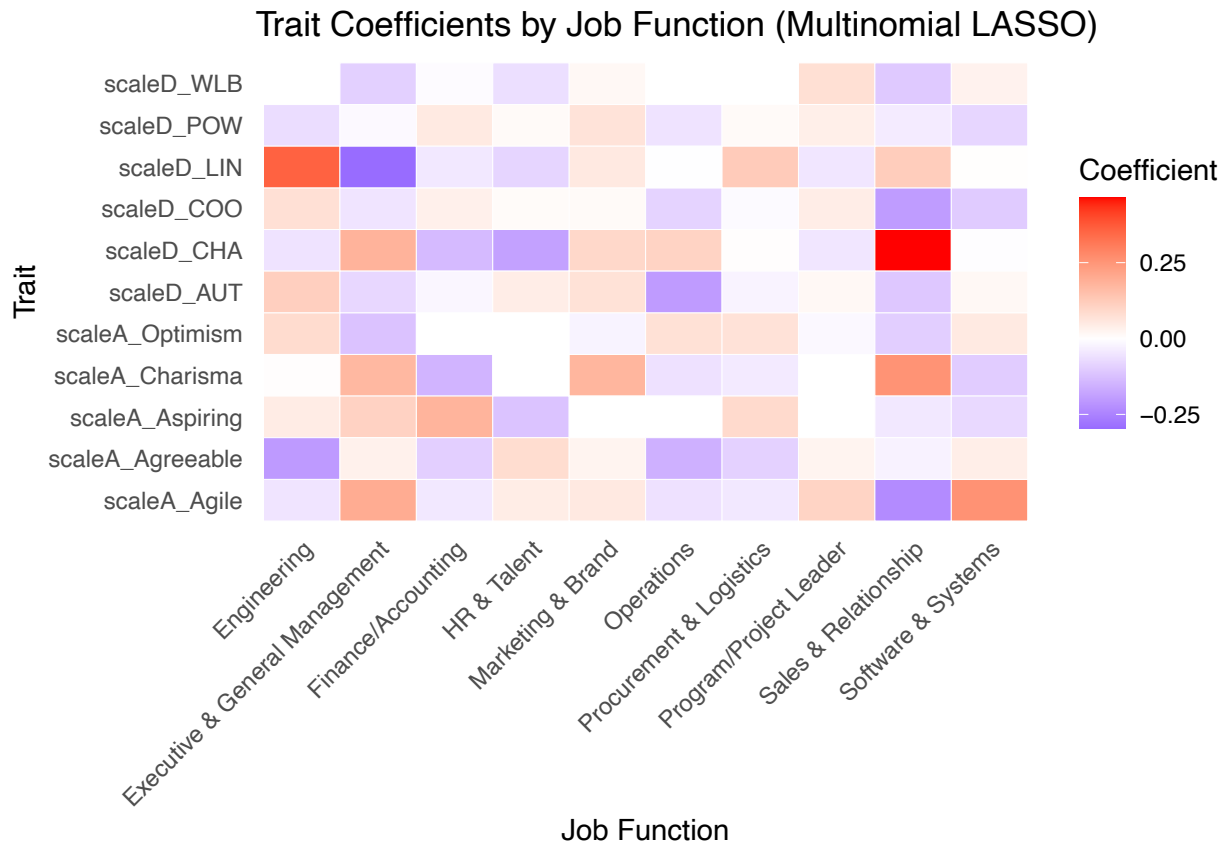
# Plot heatmap
ggplot(coef_df, aes(x = JobFunction, y = Trait, fill = Coefficient)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
  labs(

```

```

title = "Trait Coefficients by Job Function (Multinomial LASSO)",
x = "Job Function",
y = "Trait",
fill = "Coefficient"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      panel.grid = element_blank())

```



This heatmap gives us a visual summary of which psychometric traits influence the likelihood of being in each job function, based on the multinomial LASSO logistic regression.

Strong Positive Signals (Red Cells): These traits are important positive predictors for certain job functions:

scaleD_CHA (Challenge) → Sales & Relationship (bright red): - Suggests people who thrive on challenge are more likely to be in sales roles.

scaleD_LIN (Linear thinking) → Engineering: - Logical, structured thinkers are a strong fit for engineering.

scaleA_Charisma → Sales & Relationship: - Being charismatic helps in client-facing roles.

scaleA_Agile → Software & Systems: - Agile thinkers are more likely to be in software/system roles.

Strong Negative Signals (Blue Cells): These traits decrease the likelihood of being in a role:

scaleD_LIN (Linear thinking) → Executive & General Management

scaleA_Agreeable → Engineering: - Suggests engineering roles may be more independent or task-focused.

scaleA_Agile → Sales & Relationship

Focusing on Factors beside psychometric traits

```
df_nontraits <- df %>%
  filter(function. %in% (
    count(., function.) %>%
      slice_max(n, n = 10) %>%
      pull(function.)
  )) %>%
  filter(
    countryEmployed %in% (
      count(df, countryEmployed) %>%
        slice_max(n, n = 10) %>%
        pull(countryEmployed)
    ),
    countryRaised %in% (
      count(df, countryRaised) %>%
        slice_max(n, n = 10) %>%
        pull(countryRaised)
    ),
    !gender %in% c("NR", "O")
  ) %>%
  select(-starts_with("scaleA_"), -starts_with("scaleD_")) %>%
  na.omit() %>%
  mutate(across(where(is.character), as.factor))

head(df_nontraits)
```

```
##           countryEmployed      countryRaised education gender
## 1 United States of America United States of America  4-Year      M
## 2                Turkey                Turkey  Masters      M
## 3                Turkey                Turkey  4-Year      F
## 4                Turkey                Turkey  Masters      M
## 5                Turkey                Turkey  4-Year      M
## 6                Turkey                Turkey  4-Year      M
##           function. managementLevel yrsManaging yrsCurrentPosition
## 1      Software & Systems      Low-level           4              6
## 2 Executive & General Management      Executive          22              1
## 3 Executive & General Management      Executive          17              2
## 4                Operations      High-level          12              8
## 5                Engineering      Mid-level          21              8
## 6                Engineering      High-level           3              8
##    age
## 1  52
## 2  46
## 3  46
## 4  47
## 5  50
## 6  58
```

```
unique(df_nontraits$countryEmployed)
```

```
## [1] United States of America Turkey Brazil
```

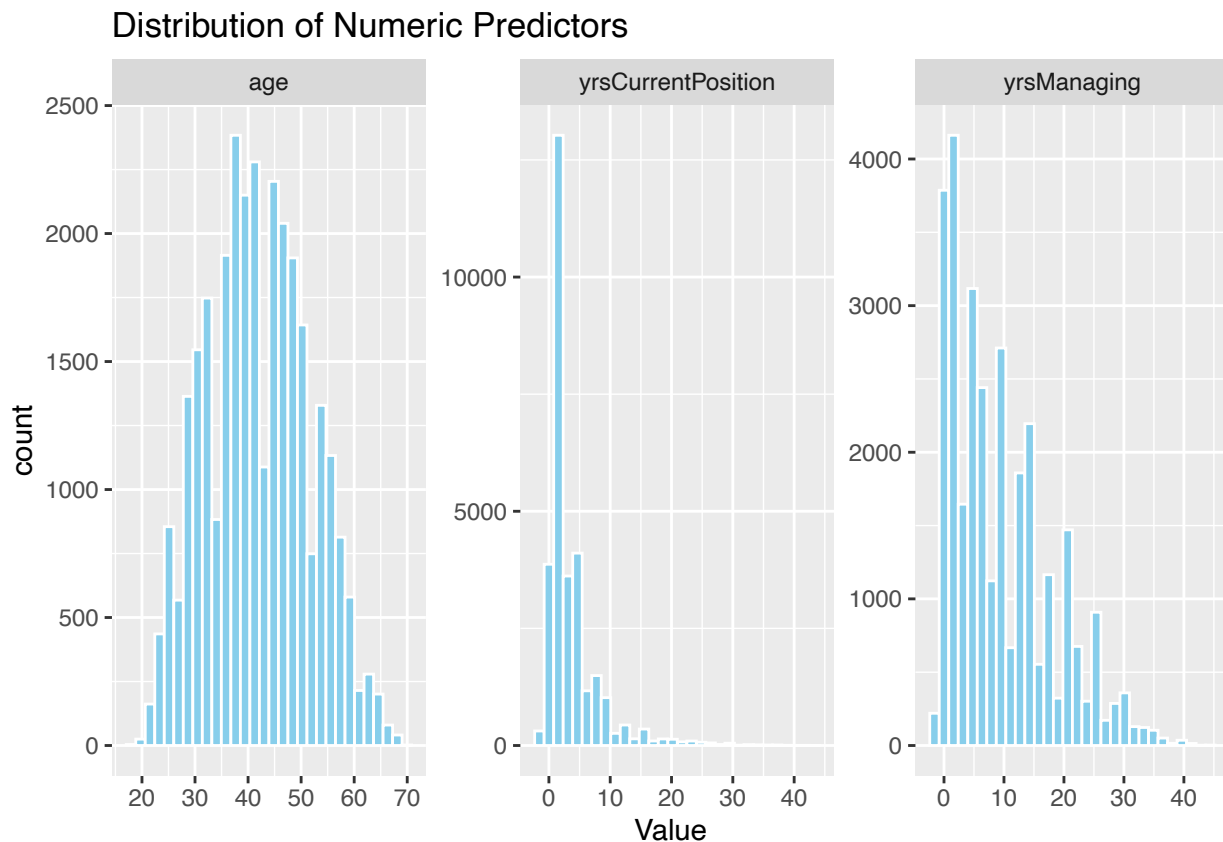
```
## [4] Australia          United Kingdom      Mexico
## [7] Sweden              India                Italy
## [10] Colombia
## 10 Levels: Australia Brazil Colombia India Italy Mexico Sweden ... United States of America
```

```
unique(df_nontraits$countryRaised)
```

```
## [1] United States of America Turkey          Brazil
## [4] United Kingdom      Australia      Mexico
## [7] Sweden              India          Italy
## [10] China
## 10 Levels: Australia Brazil China India Italy Mexico Sweden ... United States of America
```

Distributions of Numerical Predictors

```
df_nontraits %>%
  pivot_longer(where(is.numeric), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Value)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "white") +
  facet_wrap(~ Variable, scales = "free") +
  labs(title = "Distribution of Numeric Predictors")
```

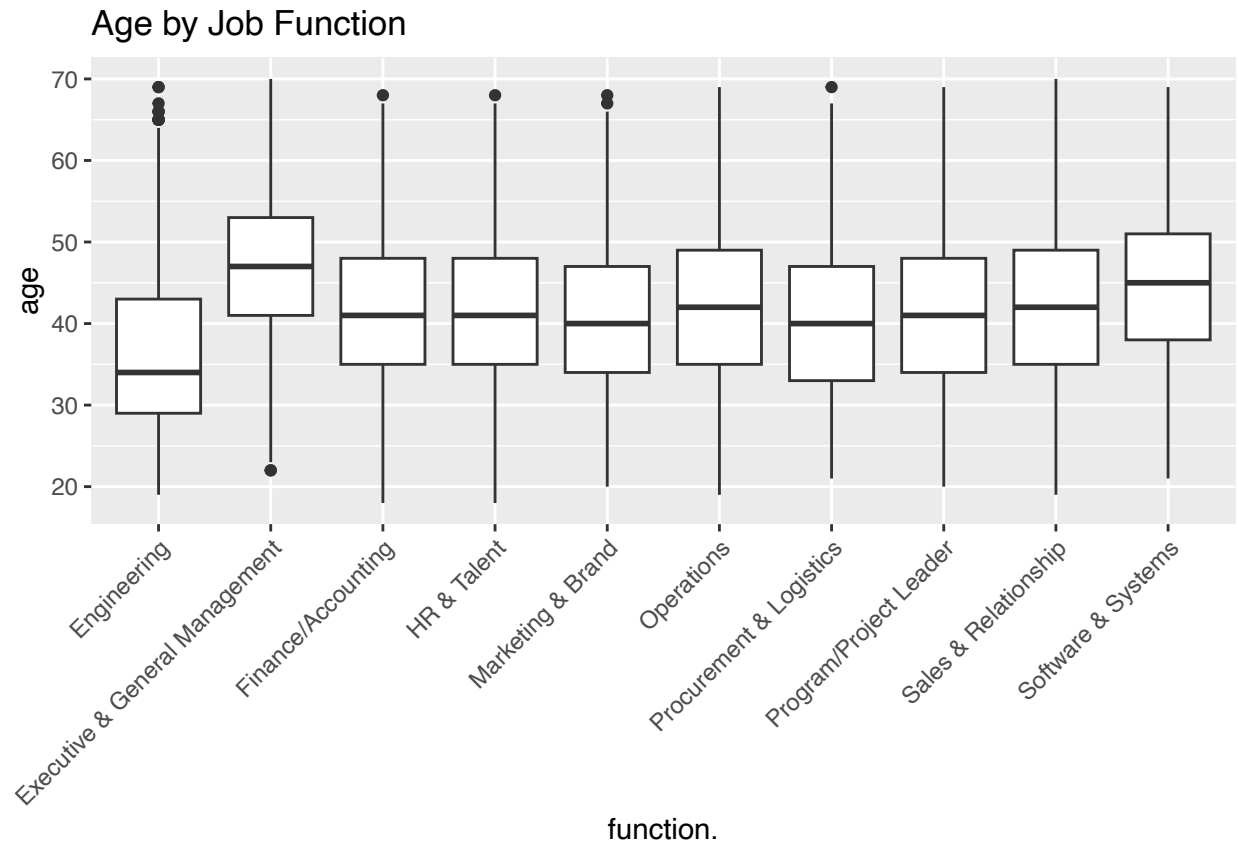


Boxplots of Numeric Predictors by Job Function

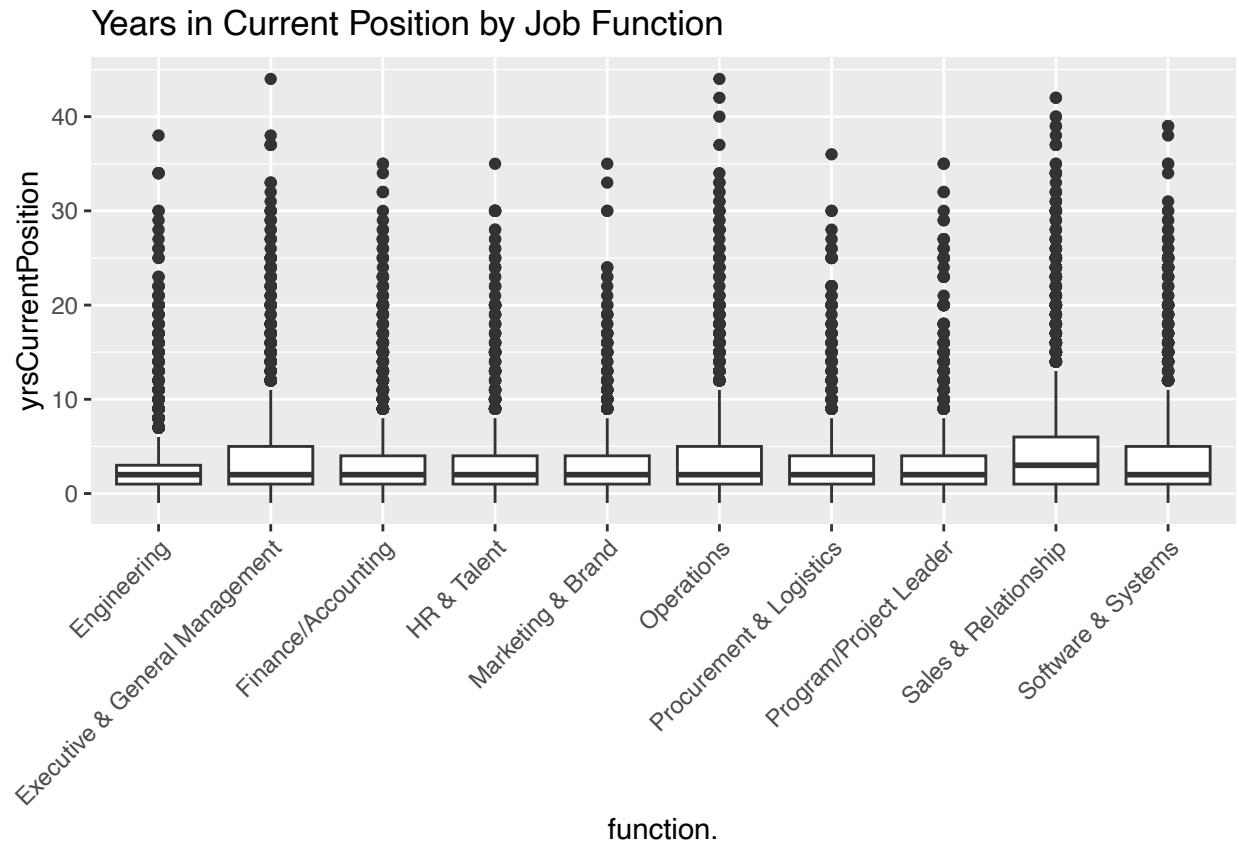
```
ggplot(df_nontraits, aes(x = function., y = yrsManaging)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(title = "Years Managing by Job Function")
```



```
ggplot(df_nontraits, aes(x = function., y = age)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(title = "Age by Job Function")
```

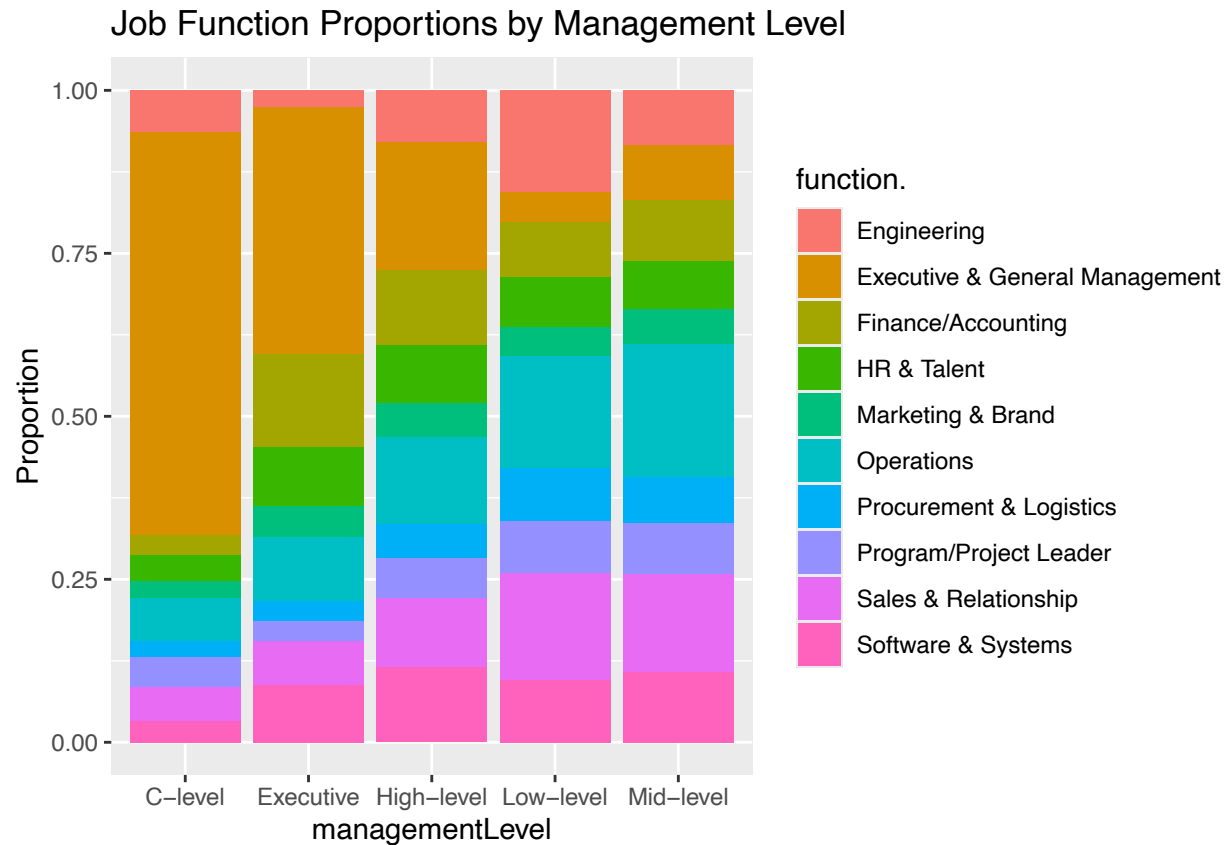


```
ggplot(df_nontraits, aes(x = function., y = yrsCurrentPosition)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(title = "Years in Current Position by Job Function")
```

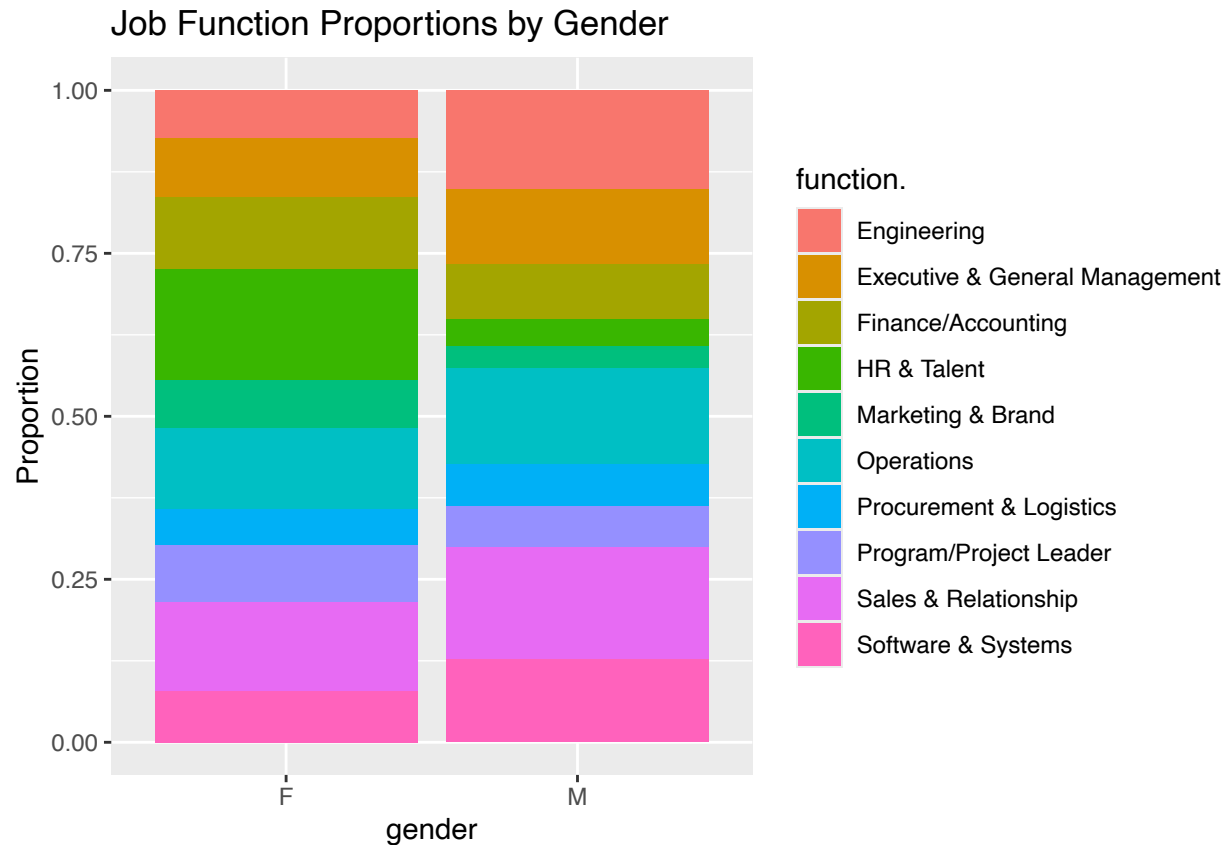


Bar Plots of Categorical Predictors

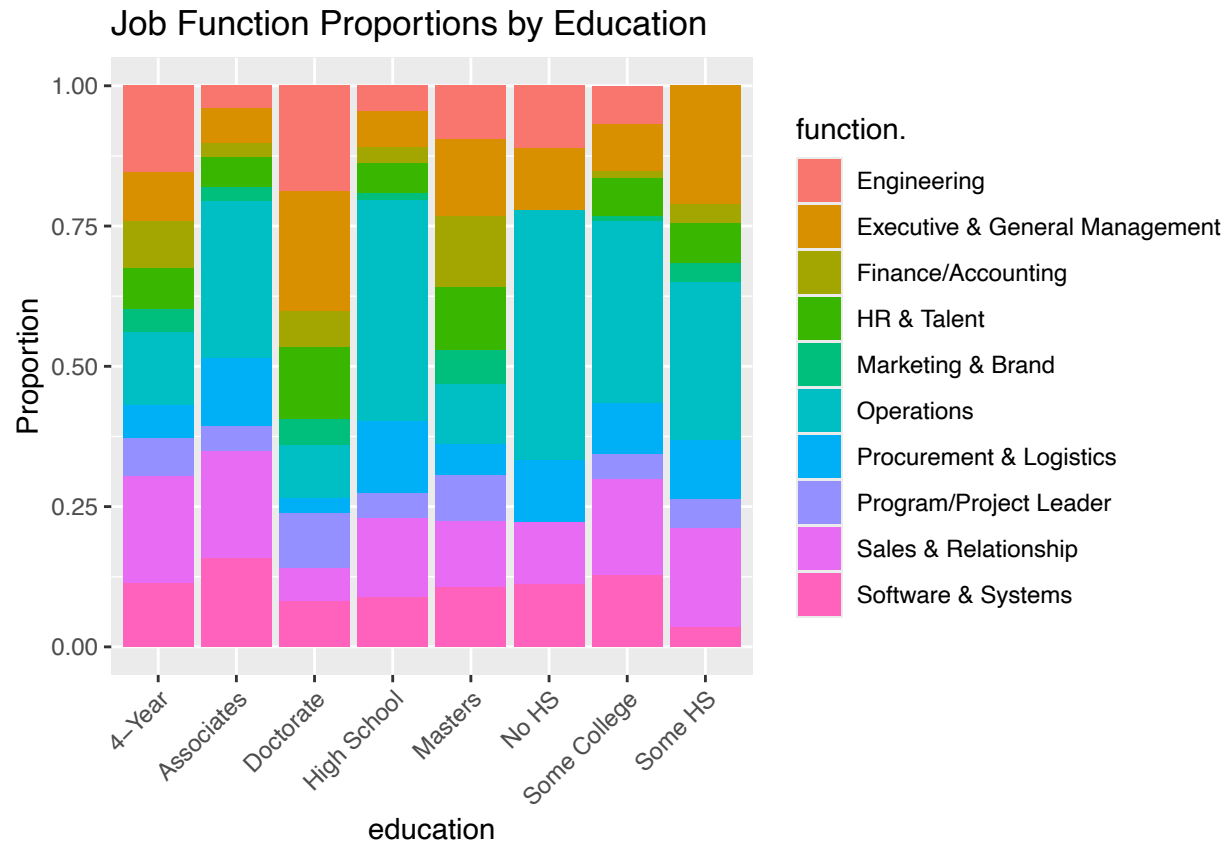
```
df_nontraits %>%
  filter(managementLevel != "None") %>%
  ggplot(aes(x = managementLevel, fill = function.)) +
  geom_bar(position = "fill") +
  labs(
    title = "Job Function Proportions by Management Level",
    y = "Proportion"
  )
```



```
ggplot(df_nontraits, aes(x = gender, fill = function.)) +
  geom_bar(position = "fill") +
  labs(title = "Job Function Proportions by Gender", y = "Proportion")
```

```
df_nontraits %>%
  filter(education != "NR") %>%
  ggplot(aes(x = education, fill = function.)) +
  geom_bar(position = "fill") +
  labs(
    title = "Job Function Proportions by Education",
    y = "Proportion"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

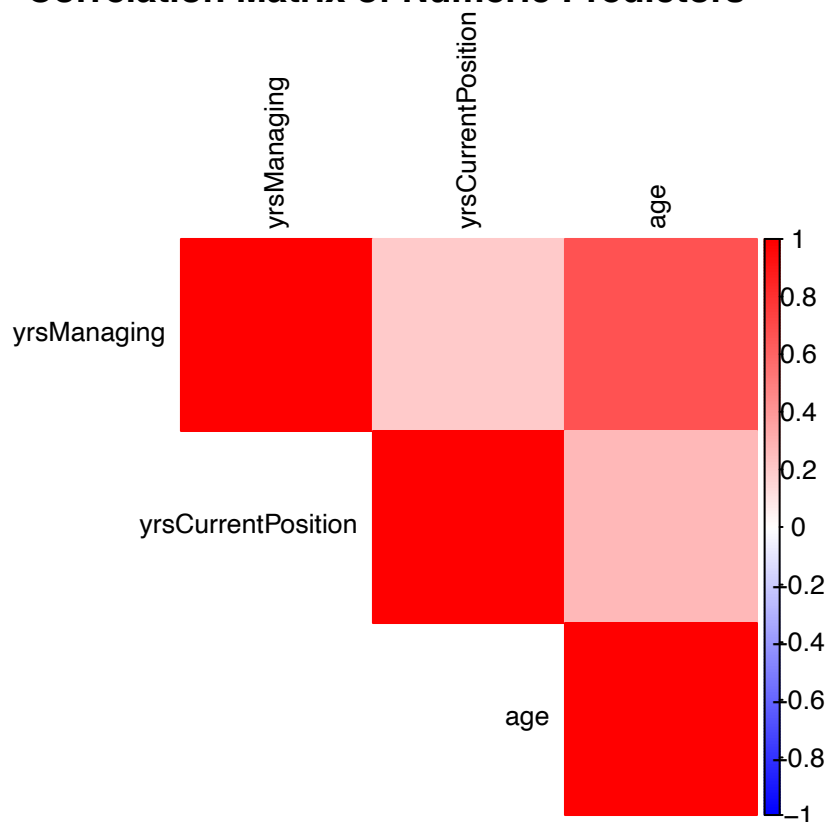


```
numeric_vars <- df_nontraits %>% select(where(is.numeric))

cor_matrix <- cor(numeric_vars, use = "complete.obs")

# correlation matrix
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  col = colorRampPalette(c("blue", "white", "red"))(200),
  tl.col = "black",
  tl.cex = 0.8,
  title = "Correlation Matrix of Numeric Predictors",
  mar = c(0, 0, 1, 0))
```

Correlation Matrix of Numeric Predictors



```
# correlation table
round(cor_matrix, 2)
```

```
##           yrsManaging yrsCurrentPosition  age
## yrsManaging           1.00           0.20 0.67
## yrsCurrentPosition      0.20           1.00 0.28
## age                    0.67           0.28 1.00
```

```
# Fit model
model <- multinom(function. ~ ., data = df_nontraits)
```

```
## # weights: 370 (324 variable)
## initial value 70512.063303
## iter 10 value 68363.494885
## iter 20 value 65646.132772
## iter 30 value 64522.866510
## iter 40 value 63417.588009
## iter 50 value 62834.934133
## iter 60 value 62270.731053
## iter 70 value 61992.432171
## iter 80 value 61774.865589
## iter 90 value 61621.227109
## iter 100 value 61476.943808
## final value 61476.943808
## stopped after 100 iterations
```

```
# Check summary
```

```
model_summary <- summary(model)
```

```
model_summary
```

```
## Call:
```

```
## multinom(formula = function. ~ ., data = df_nontraits)
```

```
##
```

```
## Coefficients:
```

```
##
## (Intercept) countryEmployedBrazil
## Executive & General Management 1.169881526 0.11978394
## Finance/Accounting 0.474893177 -1.19241021
## HR & Talent -0.002439156 -1.18866059
## Marketing & Brand -0.411376386 -0.61990896
## Operations -1.166932760 -0.31578741
## Procurement & Logistics -2.213091697 0.20102847
## Program/Project Leader -0.085848168 0.53216920
## Sales & Relationship -1.992926750 -0.91290179
## Software & Systems -3.855730700 -0.09105965
##
## countryEmployedColombia countryEmployedIndia
## Executive & General Management 1.65466368 0.663255924
## Finance/Accounting 0.05207705 -0.006478236
## HR & Talent -1.51304081 1.987794457
## Marketing & Brand 2.37909410 0.810760181
## Operations 1.17846517 1.135971130
## Procurement & Logistics -0.59254139 1.235965426
## Program/Project Leader -0.63679659 0.705210511
## Sales & Relationship -1.45106828 0.395572534
## Software & Systems -0.48606763 0.169596924
##
## countryEmployedItaly countryEmployedMexico
## Executive & General Management 0.42477025 0.04764946
## Finance/Accounting 0.02617048 -0.29007693
## HR & Talent -0.69992218 -0.21067496
## Marketing & Brand -0.96676609 0.40675393
## Operations 1.02626164 0.47733296
## Procurement & Logistics 1.49547454 1.26687160
## Program/Project Leader 1.58533893 -0.74724599
## Sales & Relationship -0.22197812 -0.23989607
## Software & Systems 0.91742468 1.08221393
##
## countryEmployedSweden countryEmployedTurkey
## Executive & General Management 0.1034709 -2.06919924
## Finance/Accounting -1.8985468 -0.11871134
## HR & Talent -1.2428320 1.32022035
## Marketing & Brand -0.9873127 -0.31902223
## Operations 1.0117140 0.02506196
## Procurement & Logistics 1.1330880 1.05701640
## Program/Project Leader 1.3289960 0.07017695
## Sales & Relationship -0.5083157 -0.83301893
## Software & Systems 0.8827475 0.49829073
##
## countryEmployedUnited Kingdom
## Executive & General Management 0.42915719
## Finance/Accounting -0.15380839
## HR & Talent -0.03452568
## Marketing & Brand 0.01817305
```

## Operations	0.66864628	
## Procurement & Logistics	0.65632738	
## Program/Project Leader	-0.30415587	
## Sales & Relationship	0.05217794	
## Software & Systems	0.28558111	
##	countryEmployedUnited States of America	
## Executive & General Management	-0.399293774	
## Finance/Accounting	-1.041781478	
## HR & Talent	-0.862973406	
## Marketing & Brand	-0.645561226	
## Operations	-0.008042887	
## Procurement & Logistics	0.284352272	
## Program/Project Leader	-0.538875187	
## Sales & Relationship	-1.163446616	
## Software & Systems	0.267556595	
##	countryRaisedBrazil countryRaisedChina	
## Executive & General Management	-0.362944728	-3.3074131
## Finance/Accounting	0.001924592	-0.6362953
## HR & Talent	0.293145479	-2.2345619
## Marketing & Brand	-0.131534102	-1.0922977
## Operations	0.758597658	-1.0577172
## Procurement & Logistics	1.267749460	0.6754469
## Program/Project Leader	-1.411678220	-1.2617976
## Sales & Relationship	0.158875025	-2.7931561
## Software & Systems	-0.237609776	-1.3595365
##	countryRaisedIndia countryRaisedItaly	
## Executive & General Management	-1.2825391	-1.3864346
## Finance/Accounting	-1.1306048	-0.8159560
## HR & Talent	-1.0948146	-0.3991906
## Marketing & Brand	-1.0597205	0.5176516
## Operations	-0.3656653	0.1504128
## Procurement & Logistics	0.6334589	0.5593635
## Program/Project Leader	-0.7890223	-1.5089173
## Sales & Relationship	-0.4981081	-0.3169195
## Software & Systems	0.6521729	-1.3459188
##	countryRaisedMexico countryRaisedSweden	
## Executive & General Management	-1.27070756	0.36477848
## Finance/Accounting	-0.63163880	-0.02296692
## HR & Talent	-0.09380228	1.53780567
## Marketing & Brand	-0.51248359	0.62501393
## Operations	0.69904012	-1.16840358
## Procurement & Logistics	0.81171977	1.02008185
## Program/Project Leader	0.12240034	-0.38342975
## Sales & Relationship	-0.16478238	-0.11231777
## Software & Systems	-1.36985843	-0.92491248
##	countryRaisedTurkey countryRaisedUnited Kingdom	
## Executive & General Management	-0.7175148	-0.2458488
## Finance/Accounting	-1.8443711	0.1934176
## HR & Talent	-2.8476251	0.8894979
## Marketing & Brand	-1.1228326	0.6083371
## Operations	-0.7115691	0.3467791
## Procurement & Logistics	-0.3416302	1.0902604
## Program/Project Leader	-1.4790093	0.5271249
## Sales & Relationship	-1.0983341	0.2751990

## Software & Systems	-1.3127507	0.3758259	
##	countryRaisedUnited States of America		
## Executive & General Management		0.08426133	
## Finance/Accounting		0.58951439	
## HR & Talent		0.96307115	
## Marketing & Brand		0.66548141	
## Operations		1.28304703	
## Procurement & Logistics		1.56992693	
## Program/Project Leader		0.58487366	
## Sales & Relationship		1.06241874	
## Software & Systems		0.74149596	
##	educationAssociates educationDoctorate		
## Executive & General Management	0.9264005	-0.03759853	
## Finance/Accounting	0.1066837	-0.84060033	
## HR & Talent	0.7712054	0.17834584	
## Marketing & Brand	0.9399390	-0.48482996	
## Operations	1.8158754	-0.84126065	
## Procurement & Logistics	1.9602220	-1.33467709	
## Program/Project Leader	0.3348415	0.04458315	
## Sales & Relationship	0.8553047	-1.53689962	
## Software & Systems	1.2434304	-1.03611344	
##	educationHigh School educationMasters		
## Executive & General Management	1.12466528	0.35488363	
## Finance/Accounting	0.03279081	0.74228956	
## HR & Talent	0.87046795	0.68338928	
## Marketing & Brand	-0.48937367	0.61812007	
## Operations	2.22154839	-0.03645844	
## Procurement & Logistics	2.14826954	0.17971368	
## Program/Project Leader	0.67287351	0.53412892	
## Sales & Relationship	0.61950446	-0.18888530	
## Software & Systems	0.71740248	0.02350711	
##	educationNo HS educationNR educationSome College		
## Executive & General Management	0.47411308	-0.2199188	0.06007413
## Finance/Accounting	-1.26304234	-0.3449738	-2.05462427
## HR & Talent	-0.92463479	-0.4591158	-0.10737093
## Marketing & Brand	-0.70516457	0.5537410	-0.39123604
## Operations	1.71272007	-0.1626569	1.22201250
## Procurement & Logistics	1.07370808	-0.1266579	0.72060788
## Program/Project Leader	-1.13168519	-0.4392848	-0.36239409
## Sales & Relationship	-0.36253329	-0.2059417	-0.04840426
## Software & Systems	0.01374509	-0.2683807	0.28631420
##	educationSome HS genderM		
## Executive & General Management	1.3351634	-0.7970789	
## Finance/Accounting	-0.3374268	-1.0674938	
## HR & Talent	-0.1130084	-2.3281265	
## Marketing & Brand	0.2460010	-1.5453570	
## Operations	1.7399561	-0.7744127	
## Procurement & Logistics	3.0506994	-0.6765169	
## Program/Project Leader	0.2277648	-1.1766247	
## Sales & Relationship	1.2617302	-0.5246428	
## Software & Systems	-0.2503367	-0.3138300	
##	managementLevelExecutive		
## Executive & General Management	0.5989634		
## Finance/Accounting	2.4100804		

## HR & Talent	1.3922206		
## Marketing & Brand	1.5104868		
## Operations	1.7497189		
## Procurement & Logistics	0.8296994		
## Program/Project Leader	0.3734191		
## Sales & Relationship	2.1553896		
## Software & Systems	2.6320605		
##	managementLevelHigh-level		
## Executive & General Management	-1.41348836		
## Finance/Accounting	0.74952457		
## HR & Talent	-0.08607622		
## Marketing & Brand	0.26994241		
## Operations	0.65221754		
## Procurement & Logistics	0.12199320		
## Program/Project Leader	-0.31226413		
## Sales & Relationship	1.18683017		
## Software & Systems	1.42381473		
##	managementLevelLow-level		
## Executive & General Management	-2.9198368		
## Finance/Accounting	0.1420714		
## HR & Talent	-0.2231592		
## Marketing & Brand	-0.1743343		
## Operations	0.5887332		
## Procurement & Logistics	0.1680814		
## Program/Project Leader	-0.2332581		
## Sales & Relationship	1.3054024		
## Software & Systems	1.0691612		
##	managementLevelMid-level	managementLevelNone	
## Executive & General Management	-2.21016642	-4.5171305	
## Finance/Accounting	0.59287601	-0.2148483	
## HR & Talent	-0.13943692	-0.1960741	
## Marketing & Brand	0.25284857	-0.4885696	
## Operations	1.03572236	-0.3769224	
## Procurement & Logistics	0.37828903	-0.4718285	
## Program/Project Leader	-0.03753619	-0.4061702	
## Sales & Relationship	1.41908644	1.5319897	
## Software & Systems	1.40532756	1.2443939	
##	yrsManaging	yrsCurrentPosition	age
## Executive & General Management	0.038430488	0.009620261	0.037039582
## Finance/Accounting	0.011808453	0.037885495	0.005175115
## HR & Talent	0.021483668	0.010502323	0.026693018
## Marketing & Brand	0.008912536	-0.009743535	0.018814240
## Operations	0.046571084	0.005247255	0.004486908
## Procurement & Logistics	0.033710430	0.001745928	0.002448928
## Program/Project Leader	0.023804277	0.002947535	0.019267631
## Sales & Relationship	-0.001370457	0.048543437	0.045269736
## Software & Systems	-0.010502598	0.019059366	0.058599123
##			
## Std. Errors:			
##	(Intercept)	countryEmployedBrazil	
## Executive & General Management	0.3144652	0.3972845	
## Finance/Accounting	0.3748719	0.3998917	
## HR & Talent	0.3650850	0.4463918	
## Marketing & Brand	0.4422073	0.5245890	

## Operations	0.3739637	0.4241190
## Procurement & Logistics	0.5108990	0.5513096
## Program/Project Leader	0.3666872	0.5906290
## Sales & Relationship	0.3970648	0.4032593
## Software & Systems	0.4481080	0.4396451
##	countryEmployedColombia	countryEmployedIndia
## Executive & General Management	1.0444054	0.2900036
## Finance/Accounting	1.3133759	0.2799997
## HR & Talent	0.9698909	0.3206367
## Marketing & Brand	1.1506066	0.3752158
## Operations	1.2919494	0.3272150
## Procurement & Logistics	0.2692436	0.4344421
## Program/Project Leader	0.6313472	0.3118684
## Sales & Relationship	0.8899128	0.2527348
## Software & Systems	0.3144876	0.2713532
##	countryEmployedItaly	countryEmployedMexico
## Executive & General Management	0.5366883	0.4303765
## Finance/Accounting	0.5193676	0.4229617
## HR & Talent	0.5942018	0.4685888
## Marketing & Brand	0.5393510	0.5368928
## Operations	0.5289240	0.4110661
## Procurement & Logistics	0.7097098	0.5497735
## Program/Project Leader	0.7174807	0.4473305
## Sales & Relationship	0.4923421	0.3969329
## Software & Systems	0.6964225	0.4888913
##	countryEmployedSweden	countryEmployedTurkey
## Executive & General Management	0.6723635	0.4659598
## Finance/Accounting	0.9752082	0.6819409
## HR & Talent	0.8013622	1.1047874
## Marketing & Brand	1.0287443	0.7509510
## Operations	0.7471603	0.5690505
## Procurement & Logistics	0.8258081	0.8484874
## Program/Project Leader	0.6222651	0.6532938
## Sales & Relationship	0.6754724	0.5783610
## Software & Systems	0.6024871	0.6336148
##	countryEmployedUnited Kingdom	
## Executive & General Management	0.3102355	
## Finance/Accounting	0.2987029	
## HR & Talent	0.3167560	
## Marketing & Brand	0.3681721	
## Operations	0.3463848	
## Procurement & Logistics	0.4654632	
## Program/Project Leader	0.3403027	
## Sales & Relationship	0.2804776	
## Software & Systems	0.3221934	
##	countryEmployedUnited States of America	
## Executive & General Management		0.2558594
## Finance/Accounting		0.2416939
## HR & Talent		0.2775315
## Marketing & Brand		0.3221646
## Operations		0.2979716
## Procurement & Logistics		0.4106755
## Program/Project Leader		0.2784264
## Sales & Relationship		0.2309737

## Software & Systems		0.2505792
##	countryRaisedBrazil	countryRaisedChina
## Executive & General Management	0.3922807	0.6432680
## Finance/Accounting	0.4007792	0.3448137
## HR & Talent	0.4494562	0.5350573
## Marketing & Brand	0.5288626	0.4889853
## Operations	0.4271923	0.4879821
## Procurement & Logistics	0.5950237	0.5720329
## Program/Project Leader	0.5943368	0.4213531
## Sales & Relationship	0.4058939	0.7231118
## Software & Systems	0.4463571	0.4454030
##	countryRaisedIndia	countryRaisedItaly
## Executive & General Management	0.2737455	0.5053419
## Finance/Accounting	0.2695391	0.5049602
## HR & Talent	0.3232973	0.5698315
## Marketing & Brand	0.3717224	0.5149814
## Operations	0.3235487	0.5167896
## Procurement & Logistics	0.4850545	0.7322051
## Program/Project Leader	0.3094647	0.7132859
## Sales & Relationship	0.2493478	0.4787201
## Software & Systems	0.2771296	0.6804847
##	countryRaisedMexico	countryRaisedSweden
## Executive & General Management	0.4051997	0.6683943
## Finance/Accounting	0.4070466	0.8856047
## HR & Talent	0.4501290	0.7759576
## Marketing & Brand	0.5267254	0.9767186
## Operations	0.3969265	0.8413211
## Procurement & Logistics	0.5799736	0.8866213
## Program/Project Leader	0.4162219	0.6750519
## Sales & Relationship	0.3800851	0.7076332
## Software & Systems	0.4781843	0.7042892
##	countryRaisedTurkey	countryRaisedUnited Kingdom
## Executive & General Management	0.4489039	0.3085477
## Finance/Accounting	0.6815617	0.3030483
## HR & Talent	1.1050131	0.3247852
## Marketing & Brand	0.7524443	0.3780769
## Operations	0.5696001	0.3551136
## Procurement & Logistics	0.8756194	0.5220979
## Program/Project Leader	0.6547243	0.3404831
## Sales & Relationship	0.5787683	0.2868987
## Software & Systems	0.6372327	0.3342534
##	countryRaisedUnited States of America	
## Executive & General Management		0.2591349
## Finance/Accounting		0.2559051
## HR & Talent		0.2952819
## Marketing & Brand		0.3432241
## Operations		0.3107423
## Procurement & Logistics		0.4766080
## Program/Project Leader		0.2965500
## Sales & Relationship		0.2451177
## Software & Systems		0.2747440
##	educationAssociates	educationDoctorate
## Executive & General Management	0.2267229	0.1452565
## Finance/Accounting	0.2579100	0.1896569

## HR & Talent	0.2256551	0.1560747
## Marketing & Brand	0.2507551	0.2156825
## Operations	0.1852894	0.1641733
## Procurement & Logistics	0.1978521	0.2692718
## Program/Project Leader	0.2475378	0.1583760
## Sales & Relationship	0.1881207	0.1850520
## Software & Systems	0.1916824	0.1652299
##	educationHigh School	educationMasters
## Executive & General Management	0.1879369	0.06139150
## Finance/Accounting	0.2242071	0.05932503
## HR & Talent	0.1881957	0.06231243
## Marketing & Brand	0.3558018	0.07299681
## Operations	0.1465765	0.05817503
## Procurement & Logistics	0.1572193	0.07011031
## Program/Project Leader	0.1926566	0.06369301
## Sales & Relationship	0.1565910	0.05561990
## Software & Systems	0.1681402	0.05885138
##	educationNo HS	educationNR educationSome College
## Executive & General Management	1.2265144	0.2741402 0.2862692
## Finance/Accounting	0.5723237	0.3006163 0.6086830
## HR & Talent	1.9716042	0.3297156 0.2947119
## Marketing & Brand	0.4062390	0.2851886 0.4005229
## Operations	0.8478686	0.2503755 0.2115342
## Procurement & Logistics	1.2561983	0.3188641 0.2646812
## Program/Project Leader	2.1125182	0.3337779 0.3239780
## Sales & Relationship	1.1881542	0.2361917 0.2236197
## Software & Systems	1.1817258	0.2589984 0.2350258
##	educationSome HS	genderM
## Executive & General Management	0.3881277	0.06159154
## Finance/Accounting	0.8225311	0.05802517
## HR & Talent	0.6840484	0.06112591
## Marketing & Brand	0.8213611	0.06918201
## Operations	0.3571591	0.05535409
## Procurement & Logistics	0.3475751	0.06597367
## Program/Project Leader	0.7355813	0.06072989
## Sales & Relationship	0.3468732	0.05328775
## Software & Systems	0.8636800	0.05944618
##	managementLevelExecutive	
## Executive & General Management	0.2740079	
## Finance/Accounting	0.3516323	
## HR & Talent	0.3317383	
## Marketing & Brand	0.3972099	
## Operations	0.3375423	
## Procurement & Logistics	0.3858544	
## Program/Project Leader	0.3493443	
## Sales & Relationship	0.3938372	
## Software & Systems	0.4184970	
##	managementLevelHigh-level	
## Executive & General Management	0.2068118	
## Finance/Accounting	0.2990400	
## HR & Talent	0.2700743	
## Marketing & Brand	0.3379030	
## Operations	0.2777719	
## Procurement & Logistics	0.3093627	

```

## Program/Project Leader 0.2687089
## Sales & Relationship 0.3404339
## Software & Systems 0.3714146
## managementLevelLow-level
## Executive & General Management 0.2110102
## Finance/Accounting 0.2992223
## HR & Talent 0.2703032
## Marketing & Brand 0.3388729
## Operations 0.2767594
## Procurement & Logistics 0.3075025
## Program/Project Leader 0.2678300
## Sales & Relationship 0.3392811
## Software & Systems 0.3714514
## managementLevelMid-level managementLevelNone
## Executive & General Management 0.2101028 0.2362282
## Finance/Accounting 0.3002099 0.3005183
## HR & Talent 0.2718808 0.2709191
## Marketing & Brand 0.3392748 0.3407231
## Operations 0.2777824 0.2793549
## Procurement & Logistics 0.3091481 0.3100413
## Program/Project Leader 0.2691175 0.2690585
## Sales & Relationship 0.3405854 0.3395705
## Software & Systems 0.3721493 0.3716483
## yrsManaging yrsCurrentPosition age
## Executive & General Management 0.005157173 0.006842055 0.004349244
## Finance/Accounting 0.005351951 0.006901770 0.004186540
## HR & Talent 0.005375216 0.007451270 0.004180211
## Marketing & Brand 0.006419381 0.009566075 0.005067579
## Operations 0.004838466 0.006551675 0.003791756
## Procurement & Logistics 0.005831066 0.007956091 0.004575844
## Program/Project Leader 0.005532858 0.007769352 0.004312307
## Sales & Relationship 0.004645007 0.005934364 0.003450918
## Software & Systems 0.004887004 0.006537457 0.003706463
##
## Residual Deviance: 122953.9
## AIC: 123601.9

```

Check significance (z-values)

```

z <- summary(model)$coefficients / summary(model)$standard.errors
p <- 2 * (1 - pnorm(abs(z)))
p

```

```

## (Intercept) countryEmployedBrazil
## Executive & General Management 1.990445e-04 0.763028106
## Finance/Accounting 2.052216e-01 0.002865286
## HR & Talent 9.946693e-01 0.007748910
## Marketing & Brand 3.522264e-01 0.237323182
## Operations 1.805785e-03 0.456530127
## Procurement & Logistics 1.479222e-05 0.715381599
## Program/Project Leader 8.148932e-01 0.367577110
## Sales & Relationship 5.190129e-07 0.023585913
## Software & Systems 0.000000e+00 0.835915520
## countryEmployedColombia countryEmployedIndia
## Executive & General Management 0.11312282 2.219228e-02

```

## Finance/Accounting	0.96837113	9.815413e-01
## HR & Talent	0.11875719	5.663461e-10
## Marketing & Brand	0.03866948	3.071205e-02
## Operations	0.36168424	5.172988e-04
## Procurement & Logistics	0.02775279	4.441861e-03
## Program/Project Leader	0.31315145	2.374414e-02
## Sales & Relationship	0.10298035	1.175434e-01
## Software & Systems	0.12220463	5.319682e-01
##	countryEmployedItaly	countryEmployedMexico
## Executive & General Management	0.42867241	0.91184176
## Finance/Accounting	0.95981231	0.49282460
## HR & Talent	0.23882853	0.65300284
## Marketing & Brand	0.07305899	0.44868607
## Operations	0.05234546	0.24555757
## Procurement & Logistics	0.03510342	0.02120289
## Program/Project Leader	0.02713356	0.09482912
## Sales & Relationship	0.65208932	0.54559476
## Software & Systems	0.18772493	0.02685573
##	countryEmployedSweden	countryEmployedTurkey
## Executive & General Management	0.87769549	8.965624e-06
## Finance/Accounting	0.05155730	8.618037e-01
## HR & Talent	0.12092584	2.320872e-01
## Marketing & Brand	0.33719310	6.709648e-01
## Operations	0.17571119	9.648712e-01
## Procurement & Logistics	0.17003354	2.128505e-01
## Program/Project Leader	0.03270068	9.144556e-01
## Sales & Relationship	0.45173027	1.497798e-01
## Software & Systems	0.14287390	4.316183e-01
##	countryEmployedUnited Kingdom	
## Executive & General Management	0.16656454	
## Finance/Accounting	0.60660822	
## HR & Talent	0.91320430	
## Marketing & Brand	0.96063224	
## Operations	0.05356265	
## Procurement & Logistics	0.15852425	
## Program/Project Leader	0.37143943	
## Sales & Relationship	0.85241928	
## Software & Systems	0.37542056	
##	countryEmployedUnited States of America	
## Executive & General Management		1.186185e-01
## Finance/Accounting		1.630075e-05
## HR & Talent		1.874287e-03
## Marketing & Brand		4.508889e-02
## Operations		9.784660e-01
## Procurement & Logistics		4.886853e-01
## Program/Project Leader		5.293738e-02
## Sales & Relationship		4.725424e-07
## Software & Systems		2.856322e-01
##	countryRaisedBrazil	countryRaisedChina
## Executive & General Management	0.35485315	2.724396e-07
## Finance/Accounting	0.99616847	6.498948e-02
## HR & Talent	0.51425761	2.962840e-05
## Marketing & Brand	0.80358411	2.549593e-02
## Operations	0.07576994	3.019425e-02

## Procurement & Logistics	0.03312323	2.376889e-01
## Program/Project Leader	0.01753869	2.747758e-03
## Sales & Relationship	0.69548678	1.121457e-04
## Software & Systems	0.59449670	2.270391e-03
##	countryRaisedIndia	countryRaisedItaly
## Executive & General Management	2.797537e-06	0.006077739
## Finance/Accounting	2.733717e-05	0.106119832
## HR & Talent	7.081562e-04	0.483589168
## Marketing & Brand	4.360410e-03	0.314807769
## Operations	2.584042e-01	0.771011307
## Procurement & Logistics	1.915682e-01	0.444900730
## Program/Project Leader	1.078356e-02	0.034392015
## Sales & Relationship	4.575531e-02	0.507962055
## Software & Systems	1.860692e-02	0.047941939
##	countryRaisedMexico	countryRaisedSweden
## Executive & General Management	0.001712673	0.58523543
## Finance/Accounting	0.120719537	0.97931030
## HR & Talent	0.834924655	0.04749977
## Marketing & Brand	0.330572377	0.52222982
## Operations	0.078216007	0.16490193
## Procurement & Logistics	0.161638982	0.24992684
## Program/Project Leader	0.768700812	0.57003468
## Sales & Relationship	0.664621936	0.87388700
## Software & Systems	0.004173939	0.18909649
##	countryRaisedTurkey	countryRaisedUnited Kingdom
## Executive & General Management	0.109960513	0.425570989
## Finance/Accounting	0.006807946	0.523317316
## HR & Talent	0.009966027	0.006167768
## Marketing & Brand	0.135634450	0.107609740
## Operations	0.211576046	0.328801837
## Procurement & Logistics	0.696419512	0.036777104
## Program/Project Leader	0.023884608	0.121581955
## Sales & Relationship	0.057734369	0.337447958
## Software & Systems	0.039390830	0.260854291
##	countryRaisedUnited States of America	
## Executive & General Management		7.450570e-01
## Finance/Accounting		2.124260e-02
## HR & Talent		1.108123e-03
## Marketing & Brand		5.251200e-02
## Operations		3.643856e-05
## Procurement & Logistics		9.878708e-04
## Program/Project Leader		4.857992e-02
## Sales & Relationship		1.462104e-05
## Software & Systems		6.957718e-03
##	educationAssociates	educationDoctorate
## Executive & General Management	4.387840e-05	7.957569e-01
## Finance/Accounting	6.791327e-01	9.326920e-06
## HR & Talent	6.316911e-04	2.531650e-01
## Marketing & Brand	1.779423e-04	2.458337e-02
## Operations	0.000000e+00	2.987644e-07
## Procurement & Logistics	0.000000e+00	7.173158e-07
## Program/Project Leader	1.761553e-01	7.783255e-01
## Sales & Relationship	5.452625e-06	0.000000e+00
## Software & Systems	8.760170e-11	3.593366e-10

##	educationHigh School	educationMasters
## Executive & General Management	2.173601e-09	7.440645e-09
## Finance/Accounting	8.837222e-01	0.000000e+00
## HR & Talent	3.739956e-06	0.000000e+00
## Marketing & Brand	1.690041e-01	0.000000e+00
## Operations	0.000000e+00	5.308542e-01
## Procurement & Logistics	0.000000e+00	1.036828e-02
## Program/Project Leader	4.783319e-04	0.000000e+00
## Sales & Relationship	7.615277e-05	6.837778e-04
## Software & Systems	1.983926e-05	6.895751e-01
##	educationNo HS	educationNR
## Executive & General Management	0.69908700	0.4224298
## Finance/Accounting	0.02732335	0.2511523
## HR & Talent	0.63908689	0.1637831
## Marketing & Brand	0.08259273	0.0521775
## Operations	0.04338027	0.5159171
## Procurement & Logistics	0.39270165	0.6912082
## Program/Project Leader	0.59216290	0.1881408
## Sales & Relationship	0.76027241	0.3832486
## Software & Systems	0.99071972	0.3000969
##	educationSome College	
## Executive & General Management	8.337833e-01	
## Finance/Accounting	7.367514e-04	
## HR & Talent	7.156153e-01	
## Marketing & Brand	3.286617e-01	
## Operations	7.608818e-09	
## Procurement & Logistics	6.478015e-03	
## Program/Project Leader	2.633210e-01	
## Sales & Relationship	8.286308e-01	
## Software & Systems	2.231387e-01	
##	educationSome HS	genderM
## Executive & General Management	5.816915e-04	0.000000e+00
## Finance/Accounting	6.816373e-01	0.000000e+00
## HR & Talent	8.687824e-01	0.000000e+00
## Marketing & Brand	7.645555e-01	0.000000e+00
## Operations	1.106670e-06	0.000000e+00
## Procurement & Logistics	0.000000e+00	0.000000e+00
## Program/Project Leader	7.568353e-01	0.000000e+00
## Sales & Relationship	2.753616e-04	0.000000e+00
## Software & Systems	7.719318e-01	1.297282e-07
##	managementLevelExecutive	
## Executive & General Management	2.882041e-02	
## Finance/Accounting	7.182477e-12	
## HR & Talent	2.707796e-05	
## Marketing & Brand	1.431035e-04	
## Operations	2.175234e-07	
## Procurement & Logistics	3.153218e-02	
## Program/Project Leader	2.851084e-01	
## Sales & Relationship	4.429958e-08	
## Software & Systems	3.188649e-10	
##	managementLevelHigh-level	
## Executive & General Management	8.220091e-12	
## Finance/Accounting	1.219552e-02	
## HR & Talent	7.499441e-01	
## Marketing & Brand	4.243626e-01	
## Operations	1.887286e-02	
## Procurement & Logistics	6.933321e-01	
## Program/Project Leader	2.451986e-01	
## Sales & Relationship	4.898834e-04	
## Software & Systems	1.263372e-04	
##	managementLevelLow-level	
## Executive & General Management	0.0000000000	
## Finance/Accounting	0.6349279893	
## HR & Talent	0.4090375111	

```
## Marketing & Brand 0.6069349380
## Operations 0.0334002752
## Procurement & Logistics 0.5846522470
## Program/Project Leader 0.3837985842
## Sales & Relationship 0.0001193028
## Software & Systems 0.0039978165
## managementLevelMid-level managementLevelNone
## Executive & General Management 0.000000e+00 0.000000e+00
## Finance/Accounting 4.828270e-02 4.746548e-01
## HR & Talent 6.080489e-01 4.692274e-01
## Marketing & Brand 4.561135e-01 1.515952e-01
## Operations 1.925939e-04 1.772534e-01
## Procurement & Logistics 2.210844e-01 1.280530e-01
## Program/Project Leader 8.890718e-01 1.311461e-01
## Sales & Relationship 3.091637e-05 6.435517e-06
## Software & Systems 1.592092e-04 8.130586e-04
## yrsManaging yrsCurrentPosition age
## Executive & General Management 9.214851e-14 1.597097e-01 0.000000e+00
## Finance/Accounting 2.735720e-02 4.036587e-08 2.164096e-01
## HR & Talent 6.420435e-05 1.586970e-01 1.707627e-10
## Marketing & Brand 1.650216e-01 3.084162e-01 2.050857e-04
## Operations 0.000000e+00 4.231880e-01 2.366774e-01
## Procurement & Logistics 7.417952e-09 8.263031e-01 5.925212e-01
## Program/Project Leader 1.689978e-05 7.044059e-01 7.893337e-06
## Sales & Relationship 7.679643e-01 2.220446e-16 0.000000e+00
## Software & Systems 3.162750e-02 3.552217e-03 0.000000e+00
```

```
top_predictors <- apply(p, 1, function(row) {
  names(sort(row, decreasing = FALSE))[1:3]
})
print(top_predictors)
```

```
## Executive & General Management Finance/Accounting
## [1,] "genderM" "educationMasters"
## [2,] "managementLevelLow-level" "genderM"
## [3,] "managementLevelMid-level" "managementLevelExecutive"
## HR & Talent Marketing & Brand Operations
## [1,] "educationMasters" "educationMasters" "educationAssociates"
## [2,] "genderM" "genderM" "educationHigh School"
## [3,] "age" "managementLevelExecutive" "genderM"
## Procurement & Logistics Program/Project Leader Sales & Relationship
## [1,] "educationAssociates" "educationMasters" "educationDoctorate"
## [2,] "educationHigh School" "genderM" "genderM"
## [3,] "educationSome HS" "age" "age"
## Software & Systems
## [1,] "(Intercept)"
## [2,] "age"
## [3,] "educationAssociates"
```

```
library(tibble)
```

```
# Get coefficients and standard errors
coefs <- model_summary$coefficients
```

```

se <- model_summary$standard.errors

# Convert to long tidy format
coef_df <- as.data.frame(coefs) %>%
  rownames_to_column("JobFunction") %>%
  pivot_longer(-JobFunction, names_to = "Trait", values_to = "Coefficient")

# Optional: also get p-values and merge
z <- coefs / se
p_vals <- 2 * (1 - pnorm(abs(z)))

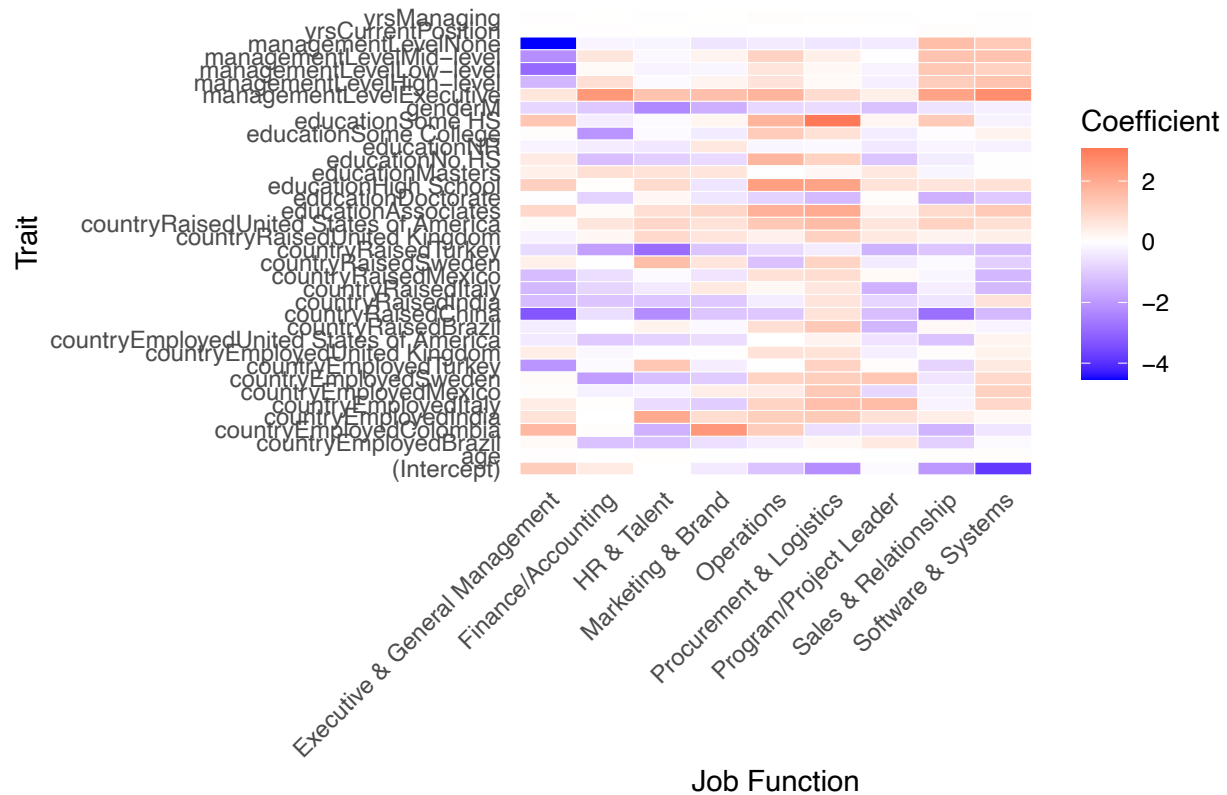
p_df <- as.data.frame(p_vals) %>%
  rownames_to_column("JobFunction") %>%
  pivot_longer(-JobFunction, names_to = "Trait", values_to = "p_value")

# Merge p-values (optional if you want to filter or annotate)
coef_df <- left_join(coef_df, p_df, by = c("JobFunction", "Trait"))

# Plot heatmap
ggplot(coef_df, aes(x = JobFunction, y = Trait, fill = Coefficient)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
  labs(
    title = "Trait Coefficients by Job Function (Multinomial Logistic Regression)",
    x = "Job Function",
    y = "Trait",
    fill = "Coefficient"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid = element_blank()
  )

```


Trait Coefficients by Job Function (Multinomial L



```
set.seed(123)

X <- model.matrix(function. ~ . - 1, data = df_nontraits) # Remove intercept
y <- df_nontraits$function.

fit <- cv.glmnet(X, y, family = "multinomial", alpha = 1)
coefs <- coef(fit, s = "lambda.min")

coefs

## $Engineering
## 37 x 1 sparse Matrix of class "dgCMatrix"
##                                     1
## (Intercept)                      0.716111548
## countryEmployedAustralia          -0.306032349
## countryEmployedBrazil              0.077885992
## countryEmployedColombia            .
## countryEmployedIndia              -1.028060262
## countryEmployedItaly               -0.210879732
## countryEmployedMexico              -0.245213470
## countryEmployedSweden              -0.467300277
## countryEmployedTurkey              .
## countryEmployedUnited Kingdom      -0.364583721
## countryEmployedUnited States of America 0.027960752
## countryRaisedBrazil                .
```

```

## countryRaisedChina          1.250495473
## countryRaisedIndia          0.661843939
## countryRaisedItaly          0.019446556
## countryRaisedMexico         0.035409163
## countryRaisedSweden         .
## countryRaisedTurkey         0.872141226
## countryRaisedUnited Kingdom -0.323247470
## countryRaisedUnited States of America -0.656842335
## educationAssociates        -0.698075061
## educationDoctorate          0.783746085
## educationHigh School        -0.761891825
## educationMasters            -0.255919294
## educationNo HS              0.419530278
## educationNR                 0.143032724
## educationSome College       -0.137281047
## educationSome HS            -1.647890975
## genderM                     0.758075433
## managementLevelExecutive     -0.763607240
## managementLevelHigh-level    -0.039192081
## managementLevelLow-level     .
## managementLevelMid-level     -0.233910911
## managementLevelNone          0.287862449
## yrsManaging                  -0.020500873
## yrsCurrentPosition           -0.007393919
## age                          -0.017463229
##
## $'Executive & General Management'
## 37 x 1 sparse Matrix of class "dgCMatrix"
##                                     1
## (Intercept)                       1.117569924
## countryEmployedAustralia          0.333807418
## countryEmployedBrazil             0.628221920
## countryEmployedColombia           1.984246216
## countryEmployedIndia              -0.006559813
## countryEmployedItaly              .
## countryEmployedMexico             -0.303084691
## countryEmployedSweden             .
## countryEmployedTurkey             -0.936732766
## countryEmployedUnited Kingdom     0.185357525
## countryEmployedUnited States of America .
## countryRaisedBrazil              .
## countryRaisedChina                -0.755263971
## countryRaisedIndia                -0.019560667
## countryRaisedItaly                -0.176311483
## countryRaisedMexico               -0.155752383
## countryRaisedSweden               0.574039387
## countryRaisedTurkey              .
## countryRaisedUnited Kingdom       .
## countryRaisedUnited States of America -0.097088345
## educationAssociates              0.088023305
## educationDoctorate               0.529469199
## educationHigh School              0.391596234
## educationMasters                  0.076346929
## educationNo HS                    0.516371633

```

```

## educationNR                                0.003166961
## educationSome College                      0.216436287
## educationSome HS                           0.709614781
## genderM                                    -0.022813465
## managementLevelExecutive                   -0.604605514
## managementLevelHigh-level                 -1.607426015
## managementLevelLow-level                  -3.058035002
## managementLevelMid-level                  -2.598452740
## managementLevelNone                       -4.096550105
## yrsManaging                               0.018389415
## yrsCurrentPosition                        0.001963025
## age                                        0.022764543
##
## $'Finance/Accounting'
## 37 x 1 sparse Matrix of class "dgCMatrix"
##                                     1
## (Intercept)                        0.1652985548
## countryEmployedAustralia            0.5298531655
## countryEmployedBrazil                -0.0219439323
## countryEmployedColombia              0.9515145740
## countryEmployedIndia                 .
## countryEmployedItaly                 0.1836530028
## countryEmployedMexico                -0.0769699402
## countryEmployedSweden                -1.1881937011
## countryEmployedTurkey               .
## countryEmployedUnited Kingdom        0.0787132823
## countryEmployedUnited States of America .
## countryRaisedBrazil                 .
## countryRaisedChina                   0.3581198397
## countryRaisedIndia                  -0.1606798175
## countryRaisedItaly                  .
## countryRaisedMexico                  .
## countryRaisedSweden                  .
## countryRaisedTurkey                  .
## countryRaisedUnited Kingdom          .
## countryRaisedUnited States of America -0.0005543606
## educationAssociates                  -0.7517023591
## educationDoctorate                   -0.1310567535
## educationHigh School                  -0.5523835917
## educationMasters                     0.3889880820
## educationNo HS                       -0.7736851035
## educationNR                          -0.0871035054
## educationSome College                 -1.2822764213
## educationSome HS                     -0.3713256951
## genderM                              -0.3370197715
## managementLevelExecutive              1.1591828038
## managementLevelHigh-level            0.5314979455
## managementLevelLow-level             .
## managementLevelMid-level             0.1087469970
## managementLevelNone                  -0.0935396678
## yrsManaging                          -0.0096635133
## yrsCurrentPosition                   0.0277233847
## age                                  -0.0058286202
##

```

```

## $'HR & Talent'
## 37 x 1 sparse Matrix of class "dgCMatrix"
##                                     1
## (Intercept)                       -0.1192565602
## countryEmployedAustralia           0.3307893041
## countryEmployedBrazil              -0.0762736494
## countryEmployedColombia            .
## countryEmployedIndia               1.3500601072
## countryEmployedItaly               -0.2708333176
## countryEmployedMexico              0.3890619056
## countryEmployedSweden              -0.1459533867
## countryEmployedTurkey              .
## countryEmployedUnited Kingdom      0.0255075940
## countryEmployedUnited States of America -0.0002453818
## countryRaisedBrazil                .
## countryRaisedChina                 -0.7716845871
## countryRaisedIndia                 -0.0105006481
## countryRaisedItaly                 -0.1508371237
## countryRaisedMexico                -0.0798554203
## countryRaisedSweden                0.4082367866
## countryRaisedTurkey                .
## countryRaisedUnited Kingdom        0.6193032613
## countryRaisedUnited States of America 0.2076081982
## educationAssociates                -0.0715495711
## educationDoctorate                 0.6768202325
## educationHigh School               0.1150003914
## educationMasters                   0.3228239864
## educationNo HS                     -0.4102897120
## educationNR                        -0.1536009416
## educationSome College              0.0402888810
## educationSome HS                   -0.0132740328
## genderM                            -1.5626524426
## managementLevelExecutive            0.4248461570
## managementLevelHigh-level          0.0174427623
## managementLevelLow-level           -0.1197818120
## managementLevelMid-level           -0.2668038214
## managementLevelNone                0.3065279016
## yrsManaging                        0.0037833595
## yrsCurrentPosition                 -0.0019630249
## age                                0.0102417402
##
## $'Marketing & Brand'
## 37 x 1 sparse Matrix of class "dgCMatrix"
##                                     1
## (Intercept)                       0.0054748978
## countryEmployedAustralia           -0.0024089364
## countryEmployedBrazil              -0.1002473968
## countryEmployedColombia            2.7553384807
## countryEmployedIndia               0.1489974442
## countryEmployedItaly               -0.0005506739
## countryEmployedMexico              0.1413035197
## countryEmployedSweden              -0.1218687690
## countryEmployedTurkey              .
## countryEmployedUnited Kingdom      0.0267726341

```

```

## countryEmployedUnited States of America .
## countryRaisedBrazil -0.0176196853
## countryRaisedChina 0.0881051171
## countryRaisedIndia -0.0538020456
## countryRaisedItaly 0.2677466029
## countryRaisedMexico .
## countryRaisedSweden .
## countryRaisedTurkey .
## countryRaisedUnited Kingdom 0.2207281337
## countryRaisedUnited States of America .
## educationAssociates -0.0875911533
## educationDoctorate 0.1310567535
## educationHigh School -0.5645351611
## educationMasters 0.2544899259
## educationNo HS .
## educationNR 0.6320299413
## educationSome College -0.8084140426
## educationSome HS .
## genderM -0.8719791084
## managementLevelExecutive 0.4606247255
## managementLevelHigh-level 0.1186841062
## managementLevelLow-level -0.2086064781
## managementLevelMid-level .
## managementLevelNone -0.2287888423
## yrsManaging -0.0021074418
## yrsCurrentPosition -0.0080153304
## age -0.0034827938
##
## $Operations
## 37 x 1 sparse Matrix of class "dgCMatrix"
## 1
## (Intercept) 0.691356489
## countryEmployedAustralia -0.854587969
## countryEmployedBrazil .
## countryEmployedColombia 1.068598128
## countryEmployedIndia .
## countryEmployedItaly 0.263733074
## countryEmployedMexico -0.002843615
## countryEmployedSweden 0.082209708
## countryEmployedTurkey -0.243346273
## countryEmployedUnited Kingdom -0.079467781
## countryEmployedUnited States of America .
## countryRaisedBrazil 0.014853779
## countryRaisedChina .
## countryRaisedIndia -0.013895812
## countryRaisedItaly 0.310142129
## countryRaisedMexico 0.439687401
## countryRaisedSweden -0.678153634
## countryRaisedTurkey .
## countryRaisedUnited Kingdom -0.165631397
## countryRaisedUnited States of America 0.166289315
## educationAssociates 1.006097567
## educationDoctorate -0.146970272
## educationHigh School 1.540945940

```

```

## educationMasters -0.294989748
## educationNo HS 1.720506247
## educationNR 0.110197816
## educationSome College 1.233392609
## educationSome HS 1.148391195
## genderM 0.022813465
## managementLevelExecutive 0.072260088
## managementLevelHigh-level .
## managementLevelLow-level 0.006569732
## managementLevelMid-level 0.150124815
## managementLevelNone -0.722535858
## yrsManaging 0.027849709
## yrsCurrentPosition 0.002400997
## age -0.013858246
##
## $'Procurement & Logistics'
## 37 x 1 sparse Matrix of class "dgCMatrix"
## 1
## (Intercept) 0.189248210
## countryEmployedAustralia -0.906464162
## countryEmployedBrazil .
## countryEmployedColombia .
## countryEmployedIndia -0.120206825
## countryEmployedItaly 0.271843743
## countryEmployedMexico 0.081901581
## countryEmployedSweden 0.381938315
## countryEmployedTurkey 0.055380481
## countryEmployedUnited Kingdom -0.025507594
## countryEmployedUnited States of America .
## countryRaisedBrazil .
## countryRaisedChina 0.563813136
## countryRaisedIndia 0.038439614
## countryRaisedItaly 0.016408086
## countryRaisedMexico 0.112114740
## countryRaisedSweden 0.060173055
## countryRaisedTurkey .
## countryRaisedUnited Kingdom -0.569851239
## countryRaisedUnited States of America -0.327818923
## educationAssociates 1.116331340
## educationDoctorate -0.451976478
## educationHigh School 1.299232828
## educationMasters -0.076346929
## educationNo HS 1.063834085
## educationNR .
## educationSome College 0.865013981
## educationSome HS 1.168724626
## genderM 0.040030388
## managementLevelExecutive -0.072260088
## managementLevelHigh-level .
## managementLevelLow-level 0.106686700
## managementLevelMid-level 0.048494886
## managementLevelNone -0.239472674
## yrsManaging 0.013790486
## yrsCurrentPosition -0.002749714

```

```

## age -0.015001541
##
## $'Program/Project Leader'
## 37 x 1 sparse Matrix of class "dgCMatrix"
## 1
## (Intercept) -0.171426439
## countryEmployedAustralia 0.002408936
## countryEmployedBrazil .
## countryEmployedColombia .
## countryEmployedIndia 0.135997697
## countryEmployedItaly 0.550969249
## countryEmployedMexico -0.180606427
## countryEmployedSweden 0.887914420
## countryEmployedTurkey -0.024795140
## countryEmployedUnited Kingdom -0.228124672
## countryEmployedUnited States of America .
## countryRaisedBrazil -0.215571720
## countryRaisedChina .
## countryRaisedIndia 0.075084375
## countryRaisedItaly .
## countryRaisedMexico .
## countryRaisedSweden .
## countryRaisedTurkey .
## countryRaisedUnited Kingdom 0.206820091
## countryRaisedUnited States of America .
## educationAssociates -0.191757788
## educationDoctorate 0.552397032
## educationHigh School .
## educationMasters 0.167488297
## educationNo HS -0.580739003
## educationNR -0.112817767
## educationSome College -0.098210732
## educationSome HS .
## genderM -0.380385470
## managementLevelExecutive -0.379488991
## managementLevelHigh-level -0.117158970
## managementLevelLow-level 0.005131151
## managementLevelMid-level -0.022623169
## managementLevelNone 0.093539668
## yrsManaging 0.002107442
## yrsCurrentPosition -0.004224818
## age 0.003482794
##
## $'Sales & Relationship'
## 37 x 1 sparse Matrix of class "dgCMatrix"
## 1
## (Intercept) -0.688890287
## countryEmployedAustralia 0.247414736
## countryEmployedBrazil .
## countryEmployedColombia .
## countryEmployedIndia 0.242690158
## countryEmployedItaly .
## countryEmployedMexico 0.211305284
## countryEmployedSweden .

```

```

## countryEmployedTurkey -0.311913711
## countryEmployedUnited Kingdom 0.116356638
## countryEmployedUnited States of America -0.199060869
## countryRaisedBrazil .
## countryRaisedChina -0.809141478
## countryRaisedIndia 0.010500648
## countryRaisedItaly -0.141547274
## countryRaisedMexico -0.131546673
## countryRaisedSweden -0.087034950
## countryRaisedTurkey -0.131586866
## countryRaisedUnited Kingdom .
## countryRaisedUnited States of America 0.104367169
## educationAssociates 0.071549571
## educationDoctorate -0.880641366
## educationHigh School -0.027495767
## educationMasters -0.448274278
## educationNo HS .
## educationNR .
## educationSome College -0.040288881
## educationSome HS 0.015986136
## genderM 0.160422415
## managementLevelExecutive -0.107252542
## managementLevelHigh-level -0.051651152
## managementLevelLow-level 0.136075628
## managementLevelMid-level 0.003283392
## managementLevelNone 0.685016438
## yrsManaging -0.015447453
## yrsCurrentPosition 0.038093115
## age 0.027651604
##
## $'Software & Systems'
## 37 x 1 sparse Matrix of class "dgCMatrix"
## 1
## (Intercept) -1.905486337
## countryEmployedAustralia -0.438382167
## countryEmployedBrazil .
## countryEmployedColombia .
## countryEmployedIndia -0.998303341
## countryEmployedItaly .
## countryEmployedMexico 0.002843615
## countryEmployedSweden 0.198990476
## countryEmployedTurkey 0.051468377
## countryEmployedUnited Kingdom -0.194325113
## countryEmployedUnited States of America 0.279113022
## countryRaisedBrazil -0.304287271
## countryRaisedChina .
## countryRaisedIndia 1.317238003
## countryRaisedItaly -0.491812119
## countryRaisedMexico -0.605247463
## countryRaisedSweden -0.588514261
## countryRaisedTurkey .
## countryRaisedUnited Kingdom .
## countryRaisedUnited States of America 0.025952872
## educationAssociates 0.384037241

```

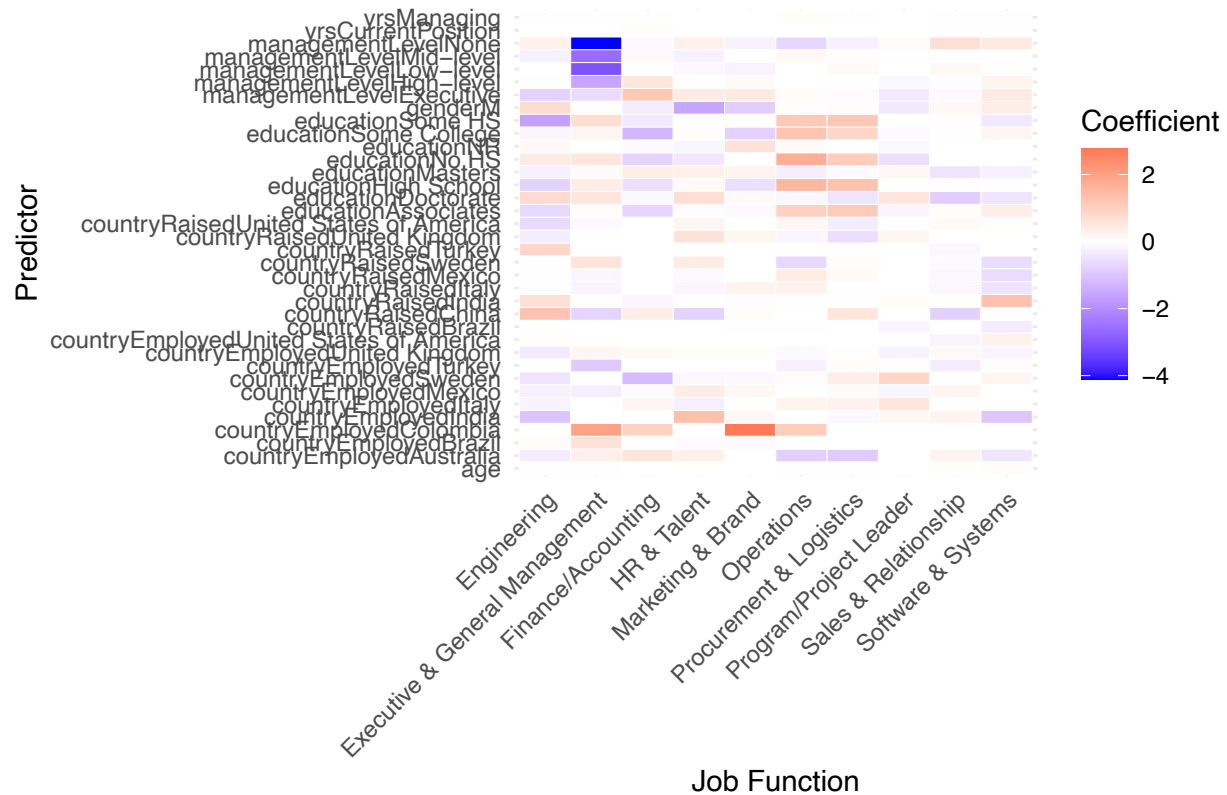


```
## educationDoctorate -0.466002653
## educationHigh School .
## educationMasters -0.239535586
## educationNo HS .
## educationNR -0.006992305
## educationSome College 0.184136271
## educationSome HS -0.393976278
## genderM 0.383756860
## managementLevelExecutive 0.440367914
## managementLevelHigh-level 0.243387575
## managementLevelLow-level -0.028105020
## managementLevelMid-level .
## managementLevelNone 0.458386936
## yrsManaging -0.026763774
## yrsCurrentPosition 0.011620945
## age 0.041607634
```

```
# Convert all coefficient matrices to a single tidy df
coef_df <- map_dfr(names(coefs), function(job) {
  mat <- as.matrix(coefs[[job]])
  data.frame(
    JobFunction = job,
    Variable = rownames(mat),
    Coefficient = mat[, 1],
    row.names = NULL
  )
}) %>%
  filter(Variable != "(Intercept)") # Remove intercepts

ggplot(coef_df, aes(x = JobFunction, y = Variable, fill = Coefficient)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0) +
  labs(
    title = "Coefficients of Non-Trait Predictors by Job Function (Multinomial LASSO)",
    x = "Job Function", y = "Predictor", fill = "Coefficient"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Coefficients of Non-Trait Predictors by Job Func



```
top3_per_job <- coef_df %>%
  filter(Coefficient != 0) %>%
  group_by(JobFunction) %>%
  slice_max(order_by = abs(Coefficient), n = 3) %>%
  ungroup()
```

```
top3_per_job
```

```
## # A tibble: 30 x 3
##   JobFunction      Variable      Coefficient
##   <chr>            <chr>            <dbl>
## 1 Engineering      educationSome HS    -1.65
## 2 Engineering      countryRaisedChina   1.25
## 3 Engineering      countryEmployedIndia -1.03
## 4 Executive & General Management managementLevelNone -4.10
## 5 Executive & General Management managementLevelLow-level -3.06
## 6 Executive & General Management managementLevelMid-level -2.60
## 7 Finance/Accounting educationSome College -1.28
## 8 Finance/Accounting countryEmployedSweden -1.19
## 9 Finance/Accounting managementLevelExecutive 1.16
## 10 HR & Talent      genderM            -1.56
## # i 20 more rows
```